

**Predicting Residential Property Values in Metro Cebu Using Machine Learning with
GIS-Derived Spatial Features**

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By

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Approval Sheet

The capstone entitled **Predicting Residential Property Values in Metro Cebu Using Machine Learning with GIS-Derived Spatial Features**, prepared and submitted by **Chris Dominic Estreba** in partial fulfillment of the requirements of the degree of **Bachelor of Science in Data Science**, is approved.

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Abstract

This study developed a residential property valuation framework for Metro Cebu using machine learning and GIS-derived spatial features. The work addressed the gap between administrative benchmarks, lender-driven appraisals, and market-facing price signals by assembling a hybrid dataset of residential properties from institutional acquired assets, foreclosure listings, and open-market online listings. The study focused on six local government units in Metro Cebu: Cebu City, Mandaue City, Lapu-Lapu City, Talisay City, Minglanilla, and Consolacion.

The methodology combined structural, locational, administrative, and geospatial variables. Property records were geocoded, matched with barangay-level BIR zonal values, and enriched with accessibility and neighborhood-context features, including proximity to major economic nodes, amenity access, and spatial lag effects. Three predictive approaches were compared: hedonic regression, Random Forest, and XGBoost. On the current post-cleanup open-market sample, the baseline Random Forest was retained as the deployed model because it delivered the strongest overall held-out performance. Model outputs were interpreted through SHAP-based explainability and prepared for deployment through a QGIS map and a Streamlit-based decision-support application.

The manuscript positions the study as a predictive and prescriptive tool for residential valuation practice in Metro Cebu. Its contribution lies in combining property-level machine learning, spatial feature engineering, and transparent explanation into a locally grounded decision-support framework that is more responsive to urban change than static administrative benchmarks alone.

Acknowledgment

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List of Maps

No maps have been listed yet.

List of Pictures

No pictures have been listed yet.

List of Appendices

Appendix A

Supporting Materials

List of Abbreviations and Acronyms

ABT	Analytics Base Table
APA	American Psychological Association
BIR	Bureau of Internal Revenue
BSP	Bangko Sentral ng Pilipinas
CBD	Central Business District
CBRT	Cebu Bus Rapid Transit
CRISP-DM	Cross-Industry Standard Process for Data Mining
GIS	Geographic Information System
IVS	International Valuation Standards
LGU	Local Government Unit
MCRAI	Metro Cebu Residential Accessibility Index
OLS	Ordinary Least Squares
OSM	OpenStreetMap
POI	Point of Interest
ROPA	Real and Other Properties Acquired
RPPI	Residential Real Estate Price Index
SHAP	SHapley Additive exPlanations

Introduction

Background of the Study

What is Real Estate?

Real estate sits at the intersection of shelter, commerce, and investment. Under Article 415 of the Philippine Civil Code, immovable property includes land, buildings, and permanent attachments. Unlike many other assets, real estate is both something people use and something they price, finance, and trade.

In actual transactions, that price is shaped by several actors with different interests. Brokers and agents facilitate sales, banks finance purchases and assess collateral risk, appraisers estimate value, and local government units set administrative benchmarks such as zonal values for taxation. Although these actors approach property from different functions, all depend on a defensible estimate of value. This study examined how that value is determined, how fair it is, and how transparent the valuation process can be.

Real Estate in Cebu

Cebu has become one of the Philippines' most dynamic regional economies, recording 7.3% growth in 2024 (PSA Region 7, 2025). Much of this growth has come from the IT-BPM sector, the expansion of call center activity, and the post-pandemic recovery of tourism, hospitality, and retail. These shifts have fed directly into demand for residential and mixed-use property.

Across Metro Cebu, particularly Cebu City, Mandaue, Lapu-Lapu, Talisay, Minglanilla, and Consolacion, that demand has made the property market notably active. Residential property prices in the area rose by 11.5% in 2025, the highest rate outside the National Capital Region (BSP, 2025).

That pressure is also being reinforced by infrastructure projects that are changing accessibility across the urban area. The Cebu Bus Rapid Transit (CBRT), the Metro Cebu Expressway, and the continued development of the South Road Properties are likely to affect land values by altering connectivity, travel times, and the attractiveness of nearby

locations. For a valuation study, these spatial shifts matter because they change not only where demand is strongest, but also how quickly existing benchmarks can become outdated.

The Problem: How is Price Decided in Cebu?

Despite this growth, the property pricing system in Metro Cebu is still assembled from several partial and often inconsistent reference points. In practice, three sources are used most often, and each has clear limitations.

First, BIR zonal values serve as administrative benchmarks for taxation, but they often lag behind actual market conditions. A 2023 appraisal review found that only 60% of LGUs nationwide had updated their assessment schedules (Otsuka et al., 2023), which suggests that some zonal values in Cebu may no longer reflect current prices. Second, bank appraisals are designed mainly to manage lending risk, so they may produce values that are more conservative than open-market prices. Third, listing prices and agent quotes often reflect asking positions rather than verified transaction values, and these are not standardized across platforms or sellers.

The result is a valuation gap between official benchmarks, estimates based on lenders, and prices that markets are seemingly willing to bear. For buyers and sellers, this makes pricing harder to interpret. For banks, it complicates collateral assessment. For local governments, it weakens the connection between tax benchmarks and market reality. Metro Cebu therefore has an active property market, but not yet a consistent and validated valuation system at the property level.

Why Metro Cebu, and Why Now?

Three converging factors make it timely to build a data-driven valuation model for Metro Cebu:

1. **Prices are moving faster than the benchmarks can follow:** With 11.5% growth in 2025 and some zonal values outdated since 2019, the gap between official

benchmarks and market reality is widening. Continued reliance on outdated models perpetuates mispricing.

2. **Infrastructure is redrawing the value map in real time:** Projects like the CBRT and the Expressway are restructuring barangay accessibility and desirability. Traditional appraisal methods, reliant on historical comparables, cannot price these spatial changes in real time. A GIS-augmented model built on geocoded property data addresses this constraint.
3. **The data now exists to do this properly:** Online platforms such as Lamudi, open geospatial data, and geocoding tools (Google Maps API) enable the construction of a property-level dataset without requiring private deed-of-sale records. Open-source GIS software also allows us to transform this data into an interactive web map, providing prescriptive spatial decision support for practitioners.

Ideal Scenario

The core problem is not that Cebu lacks valuation activity, but that the activity still lacks a common, evidence-based basis. A data-driven model that quantifies the contribution of specific value drivers, such as lot area, proximity to economic nodes, nearby amenities, and neighborhood trends, would provide reproducible estimates for residential properties. Delivered through an interactive QGIS map and a Streamlit web application, this model would help move Cebu from heuristic pricing toward a more transparent and evidence-based valuation process aligned with IVS 2025.

Statement of the Problem

Decision Problem: How can real estate firms in Metro Cebu utilize data-driven models, augmented with geospatial feature engineering, to predict property values more accurately and consistently?

Research Problem: In Metro Cebu, reliance on outdated administrative benchmarks and manual appraisals produces inconsistent, subjective pricing. No existing study combines property-level data with GIS-derived geospatial features (network distance, MCRAI scoring, spatial autocorrelation) for the Metro Cebu market, and none provides a spatial decision-support layer for valuation practice.

Research Questions

1. What value drivers significantly influence property prices in Metro Cebu?
 2. Which modeling technique—**Hedonic Regression**, **Random Forest**, or **XGBoost**—produces the most accurate valuation out of sample (lowest MAPE)?
 3. Do **geospatial features**—proximity to economic nodes, MCRAI accessibility measures, and spatial autocorrelation—significantly improve model performance compared to structural-only models?
 4. How large is the “Valuation Gap” between the model’s data-driven predictions and traditional BIR Zonal Values?
-

Significance of the Study

Addressing this valuation gap benefits multiple stakeholders:

1. **Real Estate Brokers and Appraisers:** Provides standardized tools compliant with IVS 2025 transparency requirements, including SHAP-based explainability, an interactive QGIS map for spatial analysis, and a Streamlit interface for property-level valuation review.
2. **Property Investors:** Ensures fair, data-backed pricing and helps identify undervalued areas visually through geospatial heatmaps.

3. **Banks and Lending Institutions:** Improves collateral assessment accuracy through reproducible, auditable models, which reduce exposure to pricing errors that manual appraisals can miss.
4. **Local Government Units (LGUs):** Facilitates the updating of zonal values using market-based evidence, mapping the geographic divergence between official and actual prices.

This study is significant within the Philippine context as the first Cebu-specific, data-driven valuation model integrating GIS-based feature engineering. While prior Philippine machine learning studies focus on Metro Manila or use aggregate indices, no work applies geocoded network-distance features, MCRAI scoring, and spatial autocorrelation to property-level data for Metro Cebu. By delivering results through QGIS-ready map layers and a Streamlit web application, the study extends predictive analytics into applied decision support for a fragmented local market.

Scope and Limitations

Scope

This study focuses on the residential real estate market within Metro Cebu. While the National Economic and Development Authority (NEDA) defines Metro Cebu as encompassing 13 LGUs, this study operationally scopes “Metro Cebu” to its central urban core: **Cebu City, Mandaue, Lapu-Lapu, Talisay, Minglanilla, and Consolacion**. This constraint ensures high data density and robust geospatial analysis, as these six LGUs represent the most active transaction zones and the highest concentration of infrastructure development. Figure 1 shows the spatial extent of this scope and the locations of the eight polycentric CBD nodes used in the model.

The primary data sources include:

Study Area: Metro Cebu Core Conurbation

**Figure 1**

Metro Cebu Study Area. The six LGUs in the study scope are shown alongside the eight polycentric CBD nodes used for network-distance feature engineering: Cebu Business Park, Mandaue CBD, Mactan CBD, South Road Properties, Talisay Tabunok, Consolacion, Naga City (anchor only), and Mactan-Cebu International Airport.

The study drew on six data sources covering open-market listings, bank foreclosure records, administrative floor-price benchmarks, BIR zonal values, geospatial APIs, and road network data. Each source is described in detail in Chapter 3, Section 3.2.

The study evaluated three predictive models: **Hedonic Regression (OLS)**, **Random Forest**, and **XGBoost** (see Section 1.6 for rationale). Geospatial features were engineered through geocoding, road-network distance, MCRAI scoring, and spatial lag computation. The canonical ABT preserved the wider market evidence base, but the final fitted model was restricted to the *open_market* segment so that the deployed output stayed aligned with a market-value use case.

The study produced two applied deliverables: QGIS-ready spatial layers for valuation review and a Streamlit web application combining an open-market residential price surface with property-level prediction and explanation.

Limitations

1. **Open-Market Listing Bias:** The deployed model was fitted on open-market listings rather than verified deed-of-sale transactions. This kept the prediction target aligned with market-facing pricing, but it also meant that seller strategy and asking-price noise remained in the data.
2. **Unobserved Heterogeneity:** The model cannot capture qualitative features omitted from listings, such as interior finish quality or views, typically assessed during physical inspections.
3. **Geospatial Approximation:** Addresses are geocoded via Google Maps API, which may introduce minor spatial errors compared to exact lot-parcel shapefiles. Additionally, OSM coverage in peripheral barangays may be less robust than in the urban core.
4. **Temporal Scope:** The analysis provides a cross-sectional snapshot as of late 2025; it does not capture long-term cyclical real estate trends.

5. **Geographic Coverage:** The six selected LGUs represent the urban conurbation but exclude the broader municipalities of Cebu Province.
-

Model Selection Rationale

This study employs three complementary models that span the interpretability–accuracy spectrum. This trio has been validated in comparable emerging-market real estate studies (Yacim & Boshoff, 2018; Pai & Wang, 2020).

- **Hedonic Regression (OLS)** serves as the interpretable baseline. It is the established standard in property valuation (Rosen, 1974), embedded in both the Philippine Valuation Standards (PVS, 2018) and IVS 2025. OLS produces explicit coefficients for each value driver, directly quantifying their price contribution. We extend it with a spatial lag term to account for autocorrelation among neighboring properties.
- **Random Forest** (Breiman, 2001) addresses the linearity assumption of OLS by capturing non-linear interactions between value drivers—particularly relevant for Metro Cebu housing data, where structural and locational effects are unlikely to remain linear. Its bagging mechanism also provides robustness against outliers.
- **XGBoost** (Chen & Guestrin, 2016) was included as a strong candidate for predictive accuracy: gradient boosting consistently performs well on structured tabular data (Grinsztajn et al., 2022) and integrates natively with SHAP (Lundberg & Lee, 2017) to provide property-level explainability, a practical requirement under IVS 2025 transparency standards. The final evaluation in Chapter 7 determines which model best meets the deployment criteria.

Why Not Other Models?

Several alternatives were evaluated but ultimately set aside:

Table 2*Models Considered but Not Retained in the Study Design*

Model	Rationale for Exclusion
Deep Neural Net-works	Require >10,000 labeled samples; our dataset size is insufficient, and global interpretability is sacrificed.
Support Vector Regression	Computationally expensive to tune; poor native interpretability; generally weaker than boosting on tabular data.
LASSO / Ridge	Constrained by linearity assumptions; effectively subsumed by the OLS baseline for our purposes.

Review of Related Literature

Purpose

This chapter reviews the existing body of knowledge on data-driven property valuation, drawing from both Philippine and international literature. It establishes the theoretical foundations underpinning this study, surveys empirical evidence on machine learning approaches to real estate pricing, and identifies a critical research gap: the absence of a Cebu-specific, transaction-based, ML-augmented valuation model.

The review moves from valuation concepts and the Philippine setting to the practical constraints that shape valuation work in emerging markets. It then turns to modeling approaches, the use of geospatial features in property valuation, macroeconomic influences, and the compliance issues that matter if these models are to be used in practice. The chapter closes by identifying the specific gap that this study addresses in the Metro Cebu context.

Core Concepts and Philippine Context

Valuation Foundations

Market value is the most probable price under fair, open-market conditions on the valuation date. Market price is the amount actually paid in a specific transaction; the two can differ in practice (Philippine Valuation Standards [PVS], 2018). Hedonic Pricing Theory, introduced by Rosen (1974), provides the foundational framework for this study. It posits that the price of a differentiated good—such as a house—can be decomposed into a function of its constituent attributes: $P = f(\text{Structural, Locational, Environmental})$. This theory underpins the hedonic regression model used as our interpretive baseline.

Philippine Indices and Administrative Benchmarks

The Bureau of Internal Revenue (BIR) issues zonal values primarily for tax purposes, such as capital gains tax (CGT) and documentary stamp tax (DST). These values are administratively set and are not live market prices (BIR, n.d.). Critically, the Tax Policy Study of 2023 (TPS 2023) found that only approximately 60% of Local Government Units (LGUs) have updated their assessment schedules, meaning zonal values in many areas—including parts of Cebu—are materially outdated.

For market-level context, the Bangko Sentral ng Pilipinas (BSP) publishes the Residential Real Estate Price Index (RREPI), which reported a 7.5% nationwide increase in housing prices in Q2 2025. Metro Cebu posted 11.5%—one of the highest growth rates outside the National Capital Region—reflecting sustained demand driven by IT-BPM, tourism, and infrastructure development (BSP, 2025).

Cebu-Specific Empirical Work

Agosto (2017) conducted the only Cebu-specific empirical study on land value determinants, surveying 51 real estate practitioners. The study identified transport accessibility as the primary driver of land values, followed by neighborhood quality and environmental conditions. However, the study was survey-based and did not utilize

property-level geocoded data or machine learning methods. That leaves a gap between what Cebu practitioners identify as important and what has actually been tested using property-level data and predictive models.

The Core Problem: Data Scarcity in Emerging Markets

International literature consistently identifies data scarcity—not valuer misconduct—as the primary obstacle to accurate property valuation in developing countries.

- **Kenya:** Cheloti and Mooya (2021) conducted a census survey of 427 registered valuers in Nairobi. “Limited information” was ranked as the primary valuation problem (Mean Rank 2.91), while valuer misconduct ranked last (2.32). They concluded that limited and unreliable information, rather than incompetence, causes valuation problems.
- **Nigeria (Lagos):** Ajibola (2010) surveyed 300 valuers and found that 92.7% cited insufficient market evidence as the primary challenge. Valuation inaccuracy ranged from +24.8% to +51.5%, far exceeding the global norm of $\pm 10\%$.
- **Sub-Saharan Africa:** Becsky-Nagy and Sachicola (2025) conducted a systematic review across 46 countries, finding a strong negative correlation between urbanization and credit access ($r = -0.935$). This suggested that financial infrastructure failed to keep pace with rapid urban growth—a pattern relevant to Metro Cebu’s expansion.

Taken together, these studies suggest that valuation error in emerging markets is often rooted less in professional failure than in thin, inconsistent, or inaccessible market data. That framing matters for this study because it shifts attention from individual

appraisal practice to the quality, structure, and availability of the information on which valuation depends.

Modeling Approaches: Traditional vs. Machine Learning

Hedonic Regression

Classical hedonic regression models decompose property price into a function of structural and locational attributes, typically in a log-linear form (Rosen, 1974; Malpezzi, 2003). Spatial econometrics extends this framework by incorporating neighborhood effects and spatial autocorrelation (Anselin, 1988). For the Philippines, Dann et al. (2020) applied hedonic and spatial models to Metro Manila data (2000–2010) and found that structural variables, environmental quality, and spatial spillovers all significantly explain price variation.

Philippine Machine Learning Studies

More recent Philippine work incorporates machine learning:

- **Viray (2023)** compared Multiple Linear Regression with Random Forest for predicting property prices in Central Pangasinan, combining BIR zonal values, BSP RPPI, and construction cost indices. The tree-based model achieved a lower prediction error.
- **Ramolete et al. (2023)** demonstrated that incorporating government indicators improves ML valuation accuracy, achieving a MAPE of 10–21% with larger, cleaner datasets.
- **Perdio et al. (2023)** tested linear models against gradient boosting on Manila listings, finding gradient boosting superior after feature selection.

International Evidence: The Tanzania Benchmark

The most directly relevant international study is Nyanda, Mattsson, and Wilhelmsson (2024), who tested eight ML algorithms on approximately 954 residential properties in Dar es Salaam—a sample size nearly identical to our BDO dataset.

Table 3

Dar es Salaam Benchmark Results Reported by Nyanda et al. (2024)

Model	MAPE
Neural Network	108.6% (failed due to overfitting on small data)
Random Forest	52.7%
XGBoost	48.0%

Given a sample size close to ours, the Tanzania study is useful less as a fixed benchmark than as a caution about model choice under limited data conditions. In that setting, tree-based models performed more reliably than deep learning architectures, which were more prone to overfitting. For a dataset of roughly 1,000 observations, the evidence points toward Random Forest and XGBoost as more defensible starting points than neural networks.

Southeast Asian Applications

Work from Southeast Asia points in a similar direction. In Indonesia, Wibowo et al. (2023) found that adding macroeconomic and spatial variables, including coordinates and amenity proximity, improved residential price prediction in Surabaya. In Malaysia, Samsudin et al. (2022) showed that Random Forest combined with GIS-based inputs improved valuation performance for heritage properties in Penang relative to more conventional econometric approaches. While these studies come from different property

contexts, they suggest that in fast-changing urban markets, spatial information is not just supplementary detail but part of the valuation signal itself.

Comparative Reviews

Review studies suggest that this is not an isolated pattern. Wang and Li (2020) found that although deep learning performs well when image data is available, tree-based ensembles such as Random Forest and XGBoost remain highly competitive for structured tabular datasets. Moreno-Foronda et al. (2025) and Sharma et al. (2024) likewise report strong performance from XGBoost in house price prediction tasks. Hu et al. (2024), meanwhile, show that SHAP-based explainability can narrow the interpretability gap between less transparent ML models and more conventional hedonic approaches.

Geospatial Feature Engineering in Property Valuation

For residential property, location is not a secondary attribute. It shapes accessibility, neighborhood quality, exposure to surrounding land uses, and connection to the wider urban system. Tobler's First Law of Geography, that near things tend to be more related than distant ones (Tobler, 1970), provides a clear basis for treating spatial relationships as part of the valuation problem rather than as background context.

Geocoding and Location Precision

Geocoding matters in this study because the dataset begins with imperfect addresses rather than parcel-level coordinates. Converting those addresses into latitude and longitude makes the later proximity, amenity, and neighborhood calculations possible. Services such as the Google Maps Geocoding API are widely used for this purpose because they can handle incomplete or inconsistently formatted addresses while still producing coordinates precise enough for urban spatial analysis (Google Developers, 2025). In the present study, Google Maps served as the primary geocoder for property records, while the OpenStreetMap road graph supported later routing and

network-distance calculations.

For open-source alternatives, **OpenStreetMap (OSM)** via the Nominatim geocoder provides community-maintained geospatial data at no cost. While OSM coverage varies by region, urban areas in the Philippines—particularly Metro Cebu—have beneficiary coverage from the local OpenStreetMap community and humanitarian mapping initiatives (Humanitarian OpenStreetMap Team, 2024). This study employs Google Maps API as the primary geocoding engine for address-to-coordinate conversion, while OpenStreetMap supports the road-network layer used in accessibility computation.

Proximity Analysis and Accessibility

Distance-based features—proximity to commercial centers, transportation hubs, schools, and employment nodes—are robust value drivers in hedonic pricing literature (Rosen, 1974; Malpezzi, 2003). While straight-line distance remains a common baseline in urban studies (Sinnott, 1984), current valuation workflows increasingly rely on road-network distance because it better reflects actual accessibility. This thesis follows that later approach, using network distance as the standard measure with Haversine distance only as a fallback when routing is unavailable.

Agosto (2017) confirmed that transport accessibility is the primary driver of land values in Cebu, followed by neighborhood quality. For Metro Cebu, proximity to polycentric economic nodes—including Cebu Business Park, Mandaue, Mactan, South Road Properties, Tabunok, Consolacion, Naga City, and the airport area—provides measurable value signals that are more defensible than a monocentric CBD assumption.

Accessibility Indexing and OpenStreetMap Support

Beyond direct point-to-point distances, the density and diversity of nearby amenities provides a richer description of neighborhood quality. The Python library `osmnx` enables programmatic access to OSM data for both network analysis and amenity retrieval, and has been used in property valuation research to generate proximity features from road networks and point-of-interest counts (Boeing, 2017, 2019).

In the Philippine setting, the use of OpenStreetMap cannot simply be assumed. Alvarez et al. (2021), through Project OHANA, provide more locally relevant support for its use by showing that OSM data can sustain nationwide amenity accessibility analysis in the country. Their work does not eliminate data quality concerns, but it does suggest that OSM is sufficiently robust to support spatial analysis of the kind required in this study. In the final thesis workflow, that support role is paired with external point-of-interest inventories rather than treated as the sole amenity source.

Spatial Autocorrelation and Neighbor Price Effects

A critical consideration in property valuation is **spatial autocorrelation**: properties located near each other tend to have similar prices, violating the independence assumption of standard regression (Anselin, 1988). This phenomenon is well-documented: housing prices exhibit positive spatial dependence because neighboring properties share the same schools, transportation access, environmental quality, and market conditions.

Two approaches address spatial effects in modeling:

1. **Spatial Lag Model (SLM)**: Includes a spatially weighted average of neighboring property prices as an independent variable, directly capturing the effect of nearby market conditions on a property's value.
2. **Moran's I Statistic**: A diagnostic measure of global spatial autocorrelation (Moran, 1950). A statistically significant positive Moran's I in the residuals of a non-spatial model indicates that spatial effects are present and should be incorporated.

For our study, we incorporate a **spatial lag variable**—the mean price of properties within a defined radius—as a feature in the ML models. This operationalizes Tobler's law within the predictive framework without imposing the parametric constraints of formal spatial econometric models.

Implications for Metro Cebu

This matters in Metro Cebu because value shifts are unlikely to be evenly distributed across the urban area. Infrastructure projects such as the CBRT and Metro Cebu Expressway, uneven amenity access, and strong variation across neighborhoods all create spatial differences that a purely structural model would miss. In that sense, geospatial feature engineering is not an optional enhancement but a way of making the valuation model more responsive to how the city is actually changing.

Integrating Value Drivers: Indexing vs. Raw Features

A critical challenge in geospatial ML models is how to operationalize distance metrics. Agosto (2017) identified transport accessibility and neighborhood quality as the primary value drivers in Cebu based on practitioner surveys. To integrate these findings into a machine learning pipeline, raw geospatial distances (e.g., Euclidean distance to the nearest hospital, school, or mall) are often insufficient on their own, as they can suffer from multicollinearity—a scenario where multiple distance variables are highly correlated with one another, complicating the model’s feature importance weights.

Recent work suggests that raw distance variables are often too fragmented to stand on their own. Rey-Blanco, Zofío, and González-Arias (2024), for example, show that accessibility indices built from point-of-interest data can improve housing price prediction in both hedonic regression and Random Forest models. Grouping individual distances and counts into broader measures such as transit accessibility or commercial density can preserve the locational signal while reducing overlap across variables. For this study, that approach provides a practical way to translate Agosto’s (2017) qualitative value drivers into features that can be used consistently in the model.

Macroeconomic Determinants

While structural and locational attributes shape the price of an individual property, broader macroeconomic conditions influence the direction of the market as a whole.

Nworah, Egbenta, and Ogbuefi (2023), in their study of real estate investment performance in Lagos from 2005 to 2022, found that exchange rate movements and inflation were both significantly associated with real estate outcomes.

- **Exchange Rate vs. Real Estate:** $r = -0.925$ (the strongest predictor).
- **Inflation vs. Real Estate:** $r = -0.508$ (significant but weaker).

In the Philippine case, that relationship is plausibly tied to remittance flows as well. In 2024, OFW remittances reached USD 38.3 billion nationally, and when the peso weakens, remitted income converts into more pesos. For households in Cebu, that can strengthen purchasing power and feed housing demand, which helps make sense of the 11.5% property price growth reported for Metro Cebu. The point is not that exchange rates alone explain local price movement, but that broader macro conditions can shape the direction and intensity of demand.

A macro control such as the **BSP RPPI quarterly index** is therefore relevant in principle, following Udomsap and Abid (2020), who confirmed that interest rates and macroeconomic conditions are significant determinants of housing prices.

Compliance and Explainability: IVS 2025

The International Valuation Standards (2025), effective January 31, 2025, introduced two critical new chapters relevant to data-driven valuation:

- **IVS 104 (Data and Inputs):** Mandates that data used in valuations must be Accurate, Complete, Timely, and Transparent. Sources must be traceable from their origin (“provenance requirement”).

- **IVS 105 (Valuation Models):** Explicitly states that “*No model without the valuer applying professional judgement can produce an IVS-compliant valuation.*”

Automated Valuation Models (AVMs) must be tested, transparent, and paired with professional review.

For this study, the IVS revisions matter in two practical ways.

1. **SHAP Values for Transparency:** We employ SHAP (SHapley Additive exPlanations) to provide both global feature importance (“Which value drivers affect Metro Cebu prices most?”) and local explanations (“This Mactan condominium is higher-valued because of stronger accessibility and locational advantages, but lower floor area pulls the estimate down.”). This satisfies IVS 104’s transparency requirement.
2. **Human-in-the-Loop Validation:** We engage licensed real estate brokers from the CPRE network to review model outputs, satisfying IVS 105’s professional judgement mandate. This makes our model a *decision-support tool*, not a replacement for the appraiser.

Synthesis and Research Gap

Taken together, the literature points to four recurring themes. First, valuation work in emerging markets is persistently constrained by data scarcity and uneven market information. Second, for small to mid-sized tabular datasets, tree-based machine learning models tend to perform more reliably than deep learning approaches. Third, spatial features such as proximity, accessibility measures, and neighborhood effects add information that conventional structural variables alone do not capture. Fourth, any model intended for valuation practice still has to remain transparent enough to support professional review under IVS 2025.

What remains missing is a Cebu-specific study that brings those strands together using property-level data. Prior Philippine ML studies focus on Metro Manila (Perdio et al., 2023; Dann et al., 2020), Central Pangasinan (Viray, 2023), or broader indicator-based approaches rather than property-level modeling (Ramolete et al., 2023). The only Cebu-specific study identified here, Agosto (2017), is valuable in showing what practitioners regard as important, but it is survey-based and does not test those factors using observed property data.

This thesis addresses that gap by developing a Metro Cebu valuation model that combines multi-source property evidence, tree-based machine learning, geospatial feature engineering, SHAP-based explainability, and professional review. The aim is not to replace appraisal judgement, but to produce a more transparent and spatially grounded decision-support tool for the local market.

Bridge to Methodology

Chapter 3 builds on this review by showing how those ideas are translated into data sources, engineered features, model comparisons, and evaluation procedures for Metro Cebu. In other words, the next chapter moves from what the literature suggests to how those insights are operationalized in this study.

Methodology

Research Design

This study used a quantitative, non-experimental design to model residential property values in Metro Cebu. The analysis was predictive in that it estimated prices from observed property characteristics, and prescriptive in that the results were later organized for spatial decision support through GIS. The training target across all three

models was total property price in Philippine Peso, log-transformed as *log_price* for model fitting. Price per square meter (*price_per_sqm*) served as the substantive unit-price quantity of interest and as a derived diagnostic field; evaluation results were back-transformed to total price in PHP for reporting. The independent variables included structural, geospatial, and administrative features. Macroeconomic indicators were reviewed during study design but were not retained in the final fitted specification.

Three supervised learning approaches were compared in order to balance interpretability and predictive performance, while also testing whether GIS-derived and administrative variables improved the model beyond structural features alone:

1. **Ordinary Least Squares (OLS) / Hedonic Regression:** An interpretable baseline grounded in Hedonic Pricing Theory (Rosen, 1974). Coefficients carried direct economic meaning (e.g., “each additional bedroom adds PHP X to value”).
2. **Random Forest Regressor:** A tree-based ensemble that captured non-linear relationships and feature interactions without requiring explicit specification (Breiman, 2001).
3. **XGBoost Regressor:** A gradient boosting algorithm optimized for predictive accuracy on structured tabular data (Chen & Guestrin, 2016). Empirically, XGBoost often performed well on datasets of similar scale (Nyanda et al., 2024).

Data Sources: Multi-Source Property Evidence

Because direct deed-of-sale transaction records were not publicly accessible, the broader property evidence base was assembled from multiple sources that reflected different segments of the residential market. Foreclosure and acquired-asset records were retained as conservative benchmark layers, while open-market listings provided the labels used for the deployed prediction task. The purpose of this multi-source design was not to

claim direct observation of a single true market price, but to preserve the wider valuation context while keeping the final fitted model aligned with open-market residential pricing.

Table 4

Summary of Data Sources Used in the Thesis Workflow

Source	Role	Volume	Nature
Open-Market Listings	List- Deployed training label source	Lamudi residential listings	Asking / market-facing
Bank Foreclosure Inventory	In- Conservative benchmark layer	BDO, BPI, Metrobank, Landbank, Bank of Commerce, China Bank Savings	Conservative / distressed
Floor-Price References	Refer- Conservative benchmark layer	Pag-IBIG and BDO records	Administrative / distressed
BIR Zonal Values	Administrative benchmark	Per barangay	Static / regulatory
Google Maps APIs	Geocoding and POI retrieval	Per property	Geospatial / dynamic
OpenStreetMap Road Network	Routing and network topology	Per property / destination pair	Geospatial / open data

Bank Foreclosure and Floor-Price Records

The conservative valuation layers aggregated foreclosed and acquired property listings from multiple institutional sources to prevent single-source bias:

- **BDO Unibank:** The largest single foreclosure source used in the study and the

main template for raw-to-clean pipeline design.

- **Pag-IBIG Fund (HDMF):** Acquired assets and floor-price references representing the more affordable residential segment.
- **Other bank foreclosure sources:** BPI, Metrobank, Landbank, Bank of Commerce, and China Bank Savings records were added to widen conservative market coverage.

These distressed assets were treated as a lower-bound segment of the observed market rather than as direct equivalents of open-market transaction prices. Using multiple institutional sources helped reduce the bias that might result if the dataset were drawn from only one lender or seller.

A sample of the raw BDO and bank ROPA input structure is documented in Appendix A.

Open-Market Listings

To represent current asking prices in the residential market, the study collected listings from public online platforms, primarily Lamudi. As Sousa et al. (2024) note, large listing datasets can capture pricing patterns that are often missing from sparse official records. In the final thesis workflow, these open-market listings became the training scope for the deployed price-surface model.

A sample of the cleaned Lamudi listing structure is documented in Appendix A.

Administrative and Macroeconomic Data

- **BIR Zonal Values:** Official zonal values per barangay, used as administrative benchmark features and as the basis for the derived valuation-gap diagnostic.
- **RPPI Review:** The BSP residential price index was reviewed during study design as a possible macro-control, but it was not retained in the final fitted specification. The deployed model is cross-sectional and listing-level, so quarterly RPPI movements were not joined as an active model feature in the final ABT.

Geospatial Data Sources

- **Google Maps Geocoding API:** Converted property addresses into latitude/longitude coordinates and supported reverse geocoding of barangay names for the BIR join.
- **Google Maps Places API:** Supplied the raw point-of-interest inventories that were later curated into the nine MCRAI amenity categories.
- **OpenStreetMap Road Network:** Provided the street graph used for network-distance computation through the *osmnx* workflow (Boeing, 2017, 2019). In the final implementation, OSM was used for routing and network topology rather than as the primary POI source.

Target Variable

The canonical ABT retained three market tiers—*open_market*, *bank_ropa*, and *floor_price*—to preserve the broader valuation evidence base. However, the predictive task of the thesis is the estimation of *open-market residential price*. Accordingly, model fitting filtered the ABT to *market_segment = open_market* before estimation, while foreclosure and administrative floor-price records were kept as reference layers in the wider data system rather than as training labels for the deployed model. The primary target variable for model comparison was *price_per_sqm*; total price and its log transform were retained as derived fields for diagnostics and the OLS baseline.

The complete feature matrix schema is documented in Appendix A.

Validation Layer: Human-in-the-Loop

Licensed real estate brokers from the CPRE network were included as a validation layer, not to replace statistical evaluation, but to check whether the model outputs remained plausible within local market practice. Their role was limited to two tasks:

1. **Sanity Check:** Reviewing SHAP-derived value driver rankings against practitioner knowledge.

2. **Outlier Review:** Examining cases with high prediction error to distinguish likely data issues from genuine market anomalies.
-

Data Pipeline

The data preparation process proceeded in seven stages, implemented in Python.

1. **Ingestion:** BDO acquired asset data was ingested from Excel via Pandas. Lamudi listings were collected through a custom web scraper. Pag-IBIG foreclosed property records were ingested from structured PDFs and Excel exports.
2. **Filtering:** The dataset was restricted to residential properties within Metro Cebu—Cebu City, Mandaue City, Lapu-Lapu City, Talisay City, Minglanilla, and Consolacion. The canonical ABT retained all three market tiers (*open_market*, *bank_ropa*, and *floor_price*), but the later modeling scripts filtered this ABT to *open_market* observations only before fitting.
3. **Regex Parsing and Cleaning:**
 - *BDO Data:* The *Property Description* field bundled features (e.g., “3BR 2TB”). Regex patterns extracted *Bedrooms* and *Bathrooms*.
 - *Lamudi Data:* Scraped price fields contained varying formats (e.g., “PHP 5,000,000” or “Contact agent for price”). Currency symbols and commas were stripped; rows without explicit numerical prices were dropped.
 - *Pag-IBIG Data:* Minimum bid prices were extracted and tagged as floor-price observations.
4. **Geocoding (Address to Coordinates):** Property addresses were batch-geocoded through the Google Maps Geocoding API to obtain latitude and longitude. The service was selected for project-level address resolution and reverse geocoding in

Metro Cebu, where listings often reference barangays, landmarks, or compound descriptors rather than standardized street-number formats. Results were cached locally to avoid redundant API calls.

5. **BIR Zonal Value Extraction and Barangay Join:** Official BIR zonal value schedules were sourced from four Revenue District Office files covering the six-LGU study area. A custom parser extracted street-level values by residential classification code and aggregated them to the barangay level as the median per classification. Each property was then assigned a barangay through reverse geocoding via the Google Maps Geocoding API, which served as the join key to the BIR summary table. This process yielded full `bir_zonal_rr_median` coverage across all 2,047 ABT rows.
6. **GIS Augmentation:** From geocoded coordinates, the geospatial feature set described in §3.4.1 was computed: road-network distances to eight polycentric and regional anchor nodes; nine category-specific MCRAI component scores derived from curated amenity inventories; the *mcrai_composite*; and a spatial lag variable capturing the mean price of neighboring properties within 1 km. Network shortest paths were computed on the OSM road graph, with Haversine fallback only when a point could not be snapped cleanly to the network.
7. **Final ABT Assembly:** All engineered features were merged into a single Analytics Base Table (ABT) and saved as a flat CSV for modeling. The canonical ABT preserved all market tiers, while the modeling-ready subset was produced downstream by filtering to *open_market*, dropping rows with null *price_per_sqm* or *spatial_lag_price*, and excluding three integrity-flagged price outliers.

Tools: Python (Pandas, NumPy, Scikit-learn, XGBoost, Statsmodels, Requests, OSMnx), Google Maps Geocoding and Places APIs, and QGIS for spatial visualization.

Feature Engineering

Table 5

Feature Categories Used in the Thesis Workflow

Category	Features	Source
Structural	Lot Area, Floor Area, Bedrooms, Bathrooms, Property Type, vacancy / imputation flags	BDO / Lamudi / bank records
Locational	City, Barangay, Latitude/Longitude, <i>is_mactan_island</i>	Google Maps API
Geospatial	Network distance to Cebu Business Park, Mandaue CBD, Mactan CBD, SRP, Talisay Tabunok, Consolacion, Naga City, and Airport	Geocoding + OSM road network
MCRAI	Nine component scores (education, finance, grocery, health, security, transport, tourism, recreation, retail density) plus <i>mcrai_composite</i>	Places API POI inventory + network routing
Spatial Lag	Mean price of neighboring properties within 1 km	Computed from data
Administrative	BIR zonal-value features per barangay	BIR schedules
Metadata	Source and market-segment fields retained in ABT but excluded from the final open-market feature matrix	Engineered

Engineered variables: *price_per_sqm*, *valuation_gap* (a diagnostic field, not a model feature), and derived log-price terms for the baseline regression workflow.

Geospatial Feature Engineering

Geospatial feature construction was the part of the method that most directly connected the valuation model to the urban structure of Metro Cebu. Rather than treating location as a background descriptor, this stage translated network accessibility, amenity access, and neighborhood context into variables that could enter the model explicitly.

1. **Geocoding and Administrative Join:** Property addresses were geocoded through the Google Maps Geocoding API to obtain latitude and longitude coordinates, then reverse-geocoded to recover barangay names for the BIR zonal-value join described in §3.3. Geocoding results were cached locally to preserve API quota and ensure repeatable downstream feature construction.
2. **Network Distance Features:** For each property, shortest-path road distances were computed to eight polycentric and regional anchor nodes: Cebu Business Park, Mandaue CBD, Mactan CBD, South Road Properties, Talisay Tabunok, Consolacion, Naga City, and Mactan-Cebu International Airport. This node set followed the thesis' polycentric framing of Metro Cebu and retained both internal subcenters and major external anchors that shape residential accessibility gradients (Giuliano & Small, 1991). Distances were computed on an OpenStreetMap street graph using the OSMnx workflow (Boeing, 2017, 2019). When a property or destination could not be snapped cleanly to the network, a Haversine fallback was used so that valid observations were not dropped from the ABT.
3. **Metro Cebu Residential Accessibility Index (MCRAI):** Amenity access was operationalized through the Metro Cebu Residential Accessibility Index (MCRAI), a custom gravity-style accessibility framework adapted to local valuation practice. Category-specific point inventories were assembled from Google Maps Places queries and cleaned into nine categories: education, finance, grocery, health, security, transport, tourism, recreation, and retail density. Following Hansen-style

accessibility logic, nearer opportunities contribute more strongly than farther opportunities (Hansen, 1959). For property i and category c , the category score was computed as:

$$MCRAI_{ic} = \sum_{j \in c} \frac{1}{\max(d_{ij}, 0.5)^\beta}, \quad \beta = 2,$$

where d_{ij} is the road-network distance in kilometers from property i to amenity j , floored at 0.5 km to prevent singular values for immediately adjacent points. Each category was evaluated only within its designated search radius r_c , reflecting the fact that some amenities operate at neighborhood scale while others serve wider urban catchments.

The nine *mcrai_** component scores entered the model individually. A separate *mcrai_composite* was retained as a summary index, but its final specification retained only the positive-coefficient OLS categories—education, grocery, recreation, and transport—in the composite, with weights 0.401, 0.310, 0.199, and 0.102 respectively. Security, tourism, retail density, finance, and health remained available as stand-alone feature columns, but they were not used in the composite weight vector. For the negative-coefficient categories in particular, the preferred interpretation is spatial sorting or threshold externality rather than direct residential amenity value generation (Bayer & McMillan, 2012; W. Y. Chen & Jim, 2010; Dronyk-Trosper, 2017; Song & Knaap, 2004; Tiebout, 1956; Yang et al., 2016).

4. **Spatial Lag Variable:** To capture neighborhood price effects, *spatial_lag_price* was computed as the mean price of other properties within a 1 km neighborhood of each target property. Properties with no neighbors within this radius received a null lag and were excluded from model fitting. This variable operationalized Tobler's First Law of Geography in a form that could be used directly by the non-spatial regression and tree models (Tobler, 1970).

Table 6*MCRAI Categories and Search Radii*

Category	Representative POIs	Radius (km)
Education	Schools, universities	0.8
Finance	Banks, ATMs	1.5
Grocery	Supermarkets, wet markets	2.0
Health	Hospitals, clinics	2.0
Security	Police substations, fire stations, barangay halls	2.0
Transport	Road-corridor / highway way centers	3.0
Tourism	Hotels, resorts, transient lodging	3.0
Recreation	Gyms, sports complexes, parks, beaches	1.5
Retail Density	Restaurants, cafes, bakeries, convenience stores	1.0

Pre-processing

Table 7*Pre-processing Steps Applied to the Modeling Subset*

Step	Method	Rationale
Integrity Filtering	Remove rows with null <i>price_per_sqm</i> , null <i>spatial_lag_price</i> , or explicit integrity flags	Keeps the modeling subset consistent with the final pipeline
Log Transformation	Apply log transform to the price target used by the OLS baseline	Reduces right skew while preserving the open-market target definition
Missing Values	Median imputation for remaining numeric fields; explicit zeros for structurally absent or all-null fields	Preserves usable rows while handling sparse engineered fields
Encoding	Dummy encoding for city and property type	Required for the fitted design matrix while keeping barangay for joins and reporting

Modeling Strategy

The Three Models

The rationale for selecting these three models, including alternatives that were considered but excluded, is described in Section 1.6.

Hedonic Equation

The hedonic regression took the following log-linear form for the baseline comparator:

$$\begin{aligned} \ln(\text{price_php}_i) = & \alpha + \beta_1 \ln(\text{floor area}_i) + \beta_2(\text{bedrooms}_i) + \beta_3(\text{bathrooms}_i) \\ & + \beta_4(\text{distance-to-node features}_i) + \beta_5(\text{MCRAI components}_i) \\ & + \beta_6(\text{BIR zonal features}_i) + \beta_7(\text{spatial lag}_i) + \epsilon_i \end{aligned}$$

The dependent variable is the log of total property price in Philippine Peso (*log_price*), consistent with the training target used for OLS, Random Forest, and XGBoost. Predicted values are back-transformed to total price PHP before evaluation metrics are computed. In this form, the hedonic model retains its interpretive structure while allowing administrative and spatial variables to enter directly into the price equation. This makes it possible to compare a more conventional valuation framework with the tree-based models without stripping out the locational features that are central to the study.

Hyperparameter Tuning

For the ML models (Random Forest and XGBoost), exploratory tuning was run in a separate script using **RandomizedSearchCV** with **5-fold cross-validation**. Candidate parameter combinations were sampled for both algorithms and compared against the untuned baseline specifications. The baseline Random Forest and XGBoost models were retained for the main pipeline because the tuned variants did not deliver sufficient improvement to replace the simpler benchmark configuration.

Evaluation and Explainability

Performance Metrics

Table 8
Performance Metrics Used in Model Evaluation

Metric	Description	Purpose
MAPE	Mean Absolute Percent-age Error	Primary metric; enables cross-study comparison
R ²	Coefficient of Determi-nation	Proportion of variance explained
MAE	Mean Absolute Error (in PHP)	Business-interpretable error
RMSE	Root Mean Square Error	Penalizes large errors

Benchmark Studies Used for Later Comparison

SHAP Explainability

SHAP was used to interpret model outputs at both the dataset and property level, which was important to keep the results reviewable in valuation practice and consistent with IVS 2025 transparency requirements.

- **Global SHAP (Summary Plots):** Identified which value drivers affected Metro Cebu property prices most across the entire dataset. This answered RQ1 (“What value drivers significantly influence property prices in Metro Cebu?”).
- **Local SHAP (Force Plots):** Explained individual predictions. For example: *“This Mactan condominium is predicted higher because of stronger accessibility and island-location premiums, but the smaller floor area pulls the estimate down.”*

Table 9
Benchmark Studies Used to Contextualize Later Evaluation Results

Study	Context	MAPE	Use in This Thesis
Ramolete et al. (2023)	Philippines, larger dataset	10–21%	Provides a Philippine reference point for interpreting why cleaner, larger datasets can achieve lower percentage error.
Nyanda et al. (2024)	Tanzania, $n \approx 954$	48%	Provides a small-sample emerging-market benchmark for the later evaluation chapter.

This helped keep the model interpretable enough to function as a support tool rather than a purely opaque prediction system. In practical terms, the model was evaluated not only by how accurately it predicted price, but also by whether its outputs could still be examined and discussed in terms that practitioners could understand.

Deliverables

QGIS Interactive Map

The main applied output of the study was a spatial review environment that organized model results across Metro Cebu. Rather than presenting results only as tables or summary statistics, the QGIS project was designed to show how predicted values, valuation-gap diagnostics, and locational patterns were distributed across the study area.

1. **Predicted Price Layers:** Point or surface outputs summarizing predicted open-market residential values across Metro Cebu.
2. **Valuation-Gap Diagnostics:** Comparison layers showing the divergence between model-based values and BIR zonal benchmarks.
3. **Supporting Locational Layers:** CBD-node, road-network, and amenity-accessibility context layers used to interpret the price surface geographically.

The purpose of this output was to make the model easier to inspect geographically, especially for brokers, investors, and local government users who needed to compare values across locations rather than only at the level of individual records.

Streamlit Web Application

A Streamlit web application complemented the QGIS environment by providing an interactive interface for inspecting the deployed open-market model. In practical use, the application centered on a predicted residential price surface for Metro Cebu while also supporting property-level prediction checks and SHAP-based explanation pages. While the QGIS output emphasized geographic comparison, the Streamlit application provided a more direct interface for exploring individual predictions and their underlying value drivers.

Data Understanding

This chapter covers the CRISP-DM data understanding phase for the frozen analytics base table (ABT). The goal here is simple: describe what the data looks like, where it is strong, and where it still has issues. This chapter does not discuss preprocessing or modeling steps. Those are covered in Chapter 5 and later chapters.

Initial Data Collection

The final ABT had 2,047 rows and 50 columns. It combined three segments in one file: open-market listings, bank real and other properties acquired (ROPA), and floor-price records. This setup made it easier to compare sources before applying the training filters used later in the workflow.

Table 10 shows the source mix. Lamudi had 1,619 rows (79.09%), so it dominated the file. Bank sources had 320 rows (15.63%). Most of these came from Bank of Commerce. BDO and Pag-IBIG had 108 floor-price rows (5.28%).

Table 10

Final ABT Composition by Source and Market Segment

Source	Market Segment	Rows	Share	Role in ABT
Lamudi	open_market	1,619	79.09%	Arm's-length asking-price listings used for the deployed market-value surface
Bank of Commerce	bank_ropa	234	11.28%	Distressed residential inventory
BPI	bank_ropa	65	3.13%	Distressed residential inventory
Metrobank	bank_ropa	15	0.72%	Distressed residential inventory
China Bank Savings	bank_ropa	4	0.19%	Distressed residential inventory
Landbank	bank_ropa	2	0.10%	Distressed residential inventory
PagIBIG	floor_price	96	4.63%	Administrative floor-price benchmark
BDO	floor_price	12	0.58%	Administrative floor-price benchmark
Total	all segments	2,047	100.00%	Frozen ABT before Chapter 5 filtering

The ABT covered six LGUs: Cebu City, Mandaue City, Lapu-Lapu City, Talisay City, Minglanilla, and Consolacion. Naga City stayed as a CBD distance anchor, but it was not included as a training LGU. Based on the logged decision, the four Naga listings from Phase C were dropped before merge because they were too few for stable learning.

Cebu City and Lapu-Lapu City together accounted for 66.73% of ABT rows. All 643 Lapu-Lapu rows carried *is_mactan_island* = 1. The full breakdown by LGU and market segment is shown in Appendix B.

Data Description (Schema)

The ABT columns can be grouped into eight blocks. The table below gives the main groups and their purpose.

Table 11

Analytics Base Table Schema by Feature Group

Feature Group	Representative Columns	Count	Role in Chapter 4 description
Identifiers and provenance	property_id, source, price_type, geocode_source, market_segment	7	Row identity and source provenance
Structural and flags	property_type, area_sqm, floor_area_sqm, lot_area_sqm, bedrooms, bathrooms, flag columns	10	Physical attributes and structural sparsity
Location and scope	city, latitude, longitude, barangay_geocoded, is_mactan_island	5	Study-area placement and island tagging
CBD-distance features	Eight retained dist variables for CBP, Mandaue, Mactan, SRP, Tabunok, Consolacion, Naga, and Airport	8	Network access to polycentric anchors
Administrative fields	zonal bir_zonal_rr_median, bir_zonal_cr_median, bir_zonal_rc_median, bir_zonal_rr_log	4	BIR benchmark linkage
Target and derived price fields	price_php, price_per_sqm, log_price, valuation_gap, price_outlier_flag	5	Target, transform, and diagnostics
Spatial context	spatial_lag_price	1	Neighbor-price context
MCRAI accessibility fields	Nine category scores plus mcrai_composite	10	Accessibility coverage summary
Total	Full ABT schema	50	Frozen chapter reference file

Three columns are key for this thesis flow. First, *price_per_sqm* is the target variable. Second, *valuation_gap* is a diagnostic field based on *price_per_sqm* and BIR

residential benchmarks. Third, the MCRAI block keeps both category-level scores and the `mcrai_composite` score.

This schema also shows why the ABT was not yet a final model matrix at this stage. Some fields were complete and stable, like CBD distances and most geospatial features. Other fields depended heavily on source behavior, especially structural housing fields.

Initial Data Exploration

Initial exploration focused on target distribution, property-type mix, geographic spread, and accessibility coverage. Out of 2,047 rows, 1,940 had non-null `price_per_sqm`. The 107 missing unit-price rows all came from Lamudi records with incomplete input fields needed for unit-price calculation.

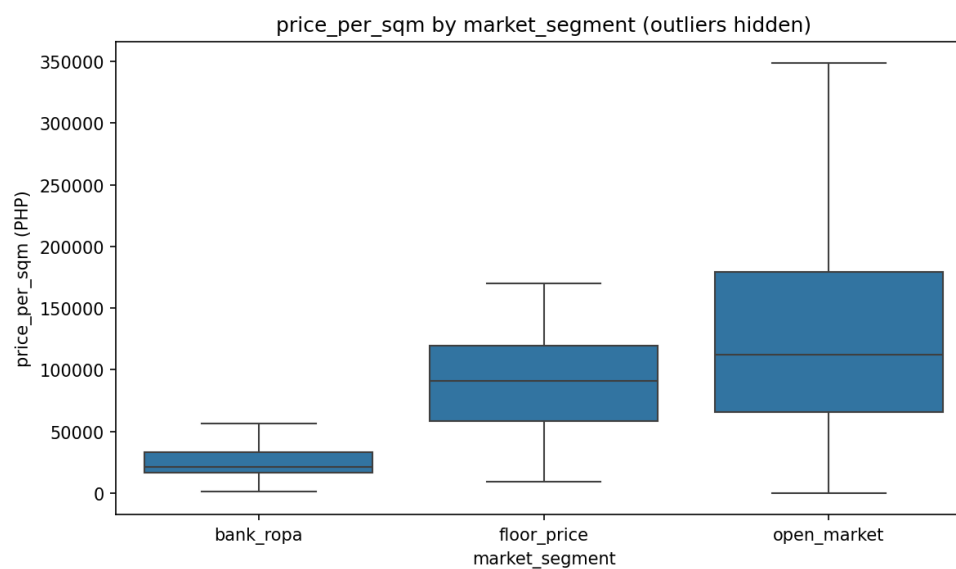
Table 12

Distribution of price_per_sqm by Market Segment

Market Segment	Non-null Rows	Q1	Median	Mean	Q3
open_market	1,512	PHP 65,323.96	PHP 111,303.56	PHP 140,135.47	PHP 178,751.00
bank_ropa	320	PHP 16,600.00	PHP 21,000.00	PHP 30,000.56	PHP 32,960.20
floor_price	108	PHP 58,506.38	PHP 91,017.27	PHP 90,188.61	PHP 119,481.21

Table 12 shows a clear spread across segments. Open-market values were much higher than bank ROPA values on both median and mean. The open-market distribution was also right-skewed, with a long upper tail.

Table 13 shows that condominiums were the largest class in the ABT. Vacant lots had much lower median unit prices than vertical housing types. This mix helped explain the wide spread in the target field.

**Figure 2**

Distribution of price_per_sqm by Market Segment. Open-market values were substantially higher than bank ROPA values, with a wider upper tail reflecting the spread in residential asking prices across Metro Cebu.

Table 13

Property-Type Composition and Median price_per_sqm

Property Type	Rows	Share	Median price_per_sqm
Condominium	862	42.11%	PHP 166,666.67
Vacant Lot	461	22.52%	PHP 18,041.24
House and Lot	412	20.13%	PHP 73,252.03
Single Detached	221	10.80%	PHP 87,500.00
Townhouse	89	4.35%	PHP 51,546.54
Apartment	2	0.10%	PHP 160,363.37

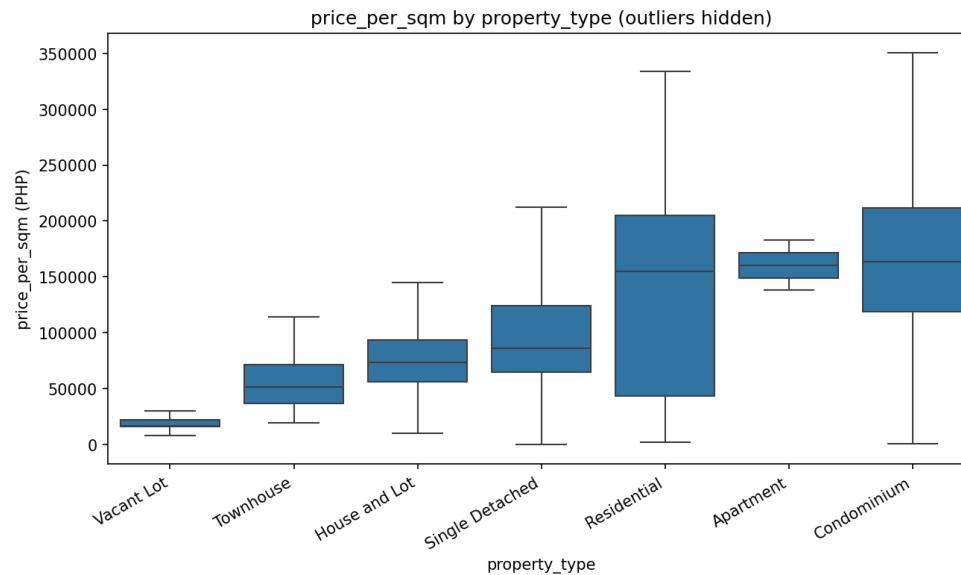


Figure 3

Median price_per_sqm by Property Type. Condominiums commanded the highest unit prices; vacant lots had the lowest, reflecting their land-only composition without built area.

Lapu-Lapu records also stood out in the geographic review. The city had 558 open-market rows, and all 643 rows from that LGU had `is_mactan_island` = 1. This confirms that a large part of the ABT sits in the island market context.

Geographic spread in the open-market subset was also uneven. Cebu City had the highest median unit price at PHP 142,886.60/sqm, followed by Lapu-Lapu City at PHP 125,799.56/sqm and Mandaue City at PHP 114,500.00/sqm. Minglanilla, Talisay City, and Consolacion formed a lower-price band, with medians between PHP 54,881.34 and PHP 70,520.26/sqm. This indicates that the premium open-market signal in the ABT remained concentrated in the core and island urban submarkets rather than the peripheral suburban LGUs. The CBD-distance matrix also showed that the southern corridor distances moved together very closely, especially the SRP–Tabunok pair ($r = 0.959$) and the Tabunok–Naga pair ($r = 0.973$). This was an early warning that the linear benchmark model would later require a collinearity check even though the nodes remained

substantively meaningful.

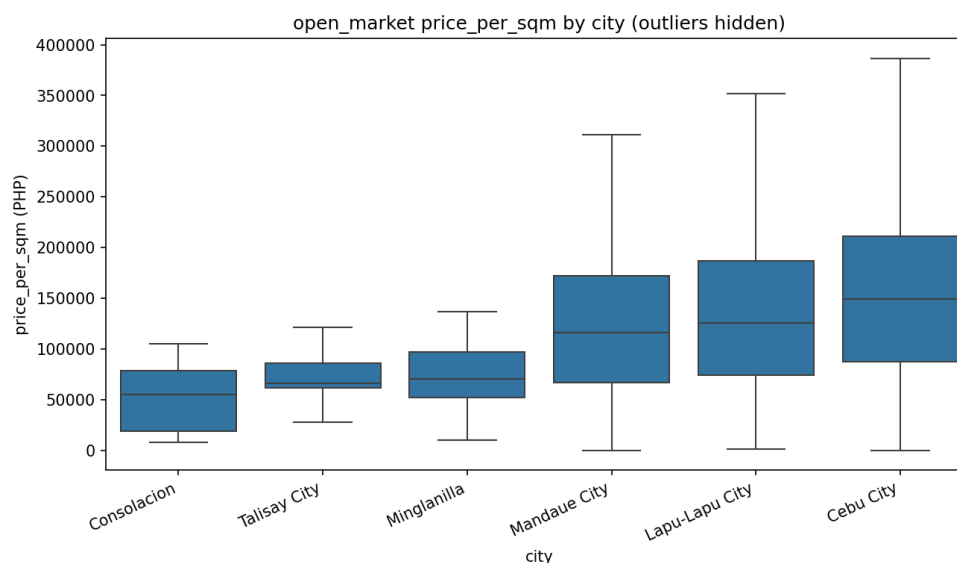


Figure 4

Median price_per_sqm by LGU for the Open-Market Subset. Cebu City and Lapu-Lapu City showed the highest median unit prices; Consolacion, Talisay, and Minglanilla formed a lower-price band.

Table 14 shows that MCRAI coverage was generally strong, but not uniform across categories. Grocery, tourism, and the composite score had very low zero rates. Education, finance, health, and security had higher zero rates.

A rank-based exploratory check on the open-market subset showed that location and accessibility fields aligned positively with unit price, while several size-related structural fields moved in the opposite direction. `bir_zonal_rr_median` ($\rho = 0.343$) and `mcrai_composite` ($\rho = 0.273$) were positively associated with `price_per_sqm`. By contrast, `floor_area_sqm` ($\rho = -0.461$), `bedrooms` ($\rho = -0.450$), and `bathrooms` ($\rho = -0.321$) were negatively associated with `price_per_sqm`. This does not mean the target was inverted. It reflects the arithmetic of a unit-price target and the composition of the sample: compact condominiums in Cebu City and Lapu-Lapu City tend to command higher PHP/sqm, while larger detached homes and land-oriented listings more often sit in

Table 14*Overall Zero Rates for Metro Cebu Residential Accessibility Index Fields*

MCRAI Field	Zero Rate	Exploratory Reading
mcrai_education	39.86%	Education access remained sparse for peripheral and island locations
mcrai_finance	36.34%	Finance nodes were still clustered around denser urban centers
mcrai_health	32.10%	Hospital and clinic access was uneven across the six-LGU scope
mcrai_security	30.22%	Security infrastructure remained absent in many residential catchments
mcrai_retail_density	21.59%	Retail and food density improved after query expansion but was not universal
mcrai_recreation	6.80%	Recreation coverage was broad but still thinner in selected peripheral locations
mcrai_transport	5.59%	Corridor-based transport access was widespread, not complete
mcrai_grocery	2.60%	Grocery access was near-universal in the final file
mcrai_tourism	0.58%	Tourism and hospitality nodes were almost always observed within the broad catchment
mcrai_composite	0.24%	Positive-weight composite coverage was effectively complete

lower-density submarkets where total asking price can still be high but price per square meter is lower. Consistent with this reading, the same structural variables remained positively associated with total asking price in the open-market subset.

The updated retail-density behavior after the targeted POI re-fetch is visible in the frozen ABT. Minglanilla had 6.74% overall zeros in `mcrai_retail_density`, and 5.63% in the open-market subset. In the same open-market subset, Talisay City had 18.82% and Lapu-Lapu City had 24.33%. For transport, open-market zero rates remained higher in Minglanilla (29.58%) and Lapu-Lapu City (14.85%).

For BIR linkage, the frozen ABT had full coverage for `bir_zonal_rr_median` across all 2,047 rows. The incomplete part was `valuation_gap`, which was missing in the same 107 rows where `price_per_sqm` was missing.

Data Quality Assessment

Data quality checks looked at missingness, integrity flags, cross-segment differences, and external-data contamination issues that were fixed before freeze.

The largest missingness issue was structural. `lot_area_sqm` was fully missing in open-market rows and heavily missing in floor-price rows. This reflects source limitations, not random noise.

The remaining structural gaps were mostly in Lamudi rows. In the frozen ABT, `floor_area_sqm` had 78 nulls total, and bedrooms and bathrooms nulls were concentrated in the open-market segment.

The ABT also kept soft outlier flags separate from hard integrity drops. There were 19 rows with `price_outlier_flag` = True (14 open-market, four bank ROPA, one floor-price). These flags did not automatically remove rows. A smaller integrity set was identified for downstream removal: row 431 (`property_id` 468), row 658 (`property_id` 714), and row 705 (`property_id` 769). Their unit prices were PHP 3.68, PHP 4.13, and PHP 4.60, which are not plausible for residential valuation in this context.

Table 15*Major Missingness Patterns in the Final ABT*

Field	Nulls	Share	Concentration	Reading for Chapter 5
lot_area_sqm	1,758	84.72%	100.00% of open_market rows; 82.41% of floor_price rows	Structural source absence, not random missingness
barangay_geocoded	834	40.19%	Weakest barangay recovery in Consolacion and Mandaue	Location granularity remained uneven
bedrooms	334	16.10%	20.28% of open_market rows	Mostly Lamudi field gaps
bathrooms	237	11.42%	14.39% of open_market rows	Mostly Lamudi field gaps
price_per_sqm	107	5.23%	All 107 nulls came from Lamudi rows	Target derivation incomplete for a small open-market subset
valuation_gap	107	5.23%	Same rows as price_per_sqm	Diagnostic field depends on target availability
floor_area_sqm	78	3.76%	4.74% of open_market rows	Residual Lamudi incompleteness
area_sqm and log_price	78 each	3.76%	Same rows as floor_area_sqm nulls	Derived-field propagation of source gaps
spatial_lag_price	23	1.11%	Thinly spread across all segments	Neighbor-based context unavailable for a small edge subset

Cross-segment behavior was also clear. Bank ROPA unit prices were systematically lower than open-market values. Floor-price records represented a different pricing basis. This supports the later segmentation step tied to valuation standards, where open_market is used as the modeling scope for Market Value estimation (International Valuation Standards Council, 2025).

Another issue was fixed before freeze in the POI pipeline. Earlier runs had a bounding-box bug that allowed out-of-scope points (for example from Bohol, Batangas, and Tagaytay) to pass the LGU filter. The corrected freeze removed this contamination

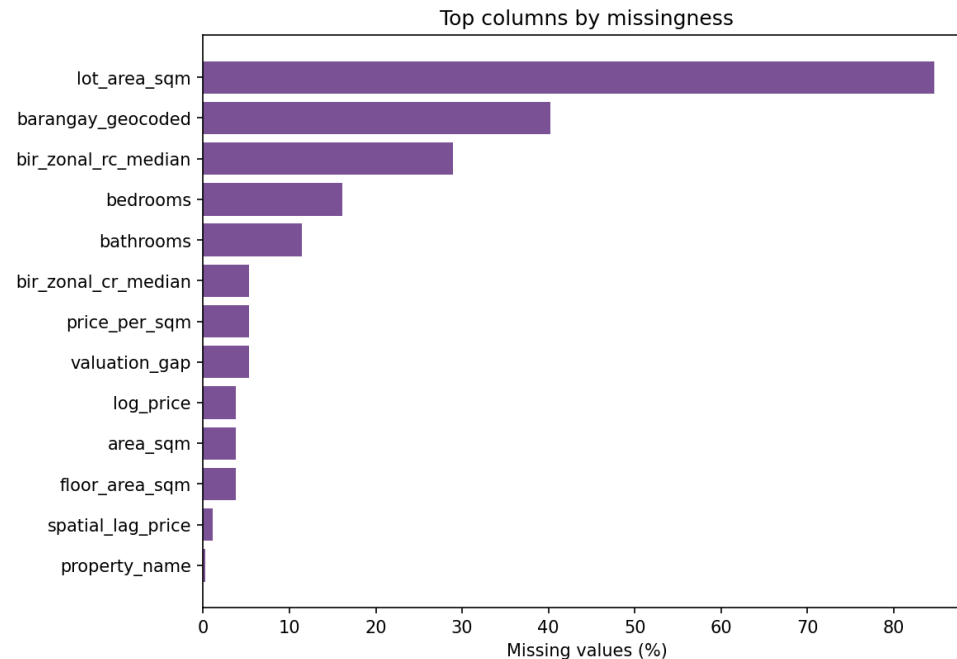


Figure 5

Top-15 Fields by Missingness Rate in the Frozen ABT. lot_area_sqm dominated because it was structurally absent in virtually all open-market rows rather than randomly missing.

before final MCRAI recomputation.

Overall, the ABT at this stage was rich and usable for analysis, but still not model-ready in raw form. It had strong geospatial and benchmark coverage, but it also had source-driven structural gaps, a few implausible rows, and clear segment differences in price behavior. Chapter 5 explains how these findings were handled in data preparation.

Data Preparation

This chapter covers the CRISP-DM data preparation phase for the frozen analytics base table (ABT). Chapter 4 described the condition of the data. This chapter explains what was done to turn that file into a modeling-ready dataset. The focus here is on filtering, missing-data treatment, feature engineering, encoding, and the final train-test split used by the modeling script.

Scope of Preparation Work

The preparation work started from the frozen ABT with 2,047 rows and 50 columns. The ABT already reflected the six-LGU study scope used in this thesis: Cebu City, Mandaue City, Lapu-Lapu City, Talisay City, Minglanilla, and Consolacion. Naga City stayed in the system only as a CBD distance anchor through `dist_naga_city_m`. Its four scraped listings were excluded before the ABT freeze because they were too few to support stable LGU-level learning.

The goal of this phase was more specific than general ABT cleaning. It was to build the exact dataset used by the modeling pipeline in `thesis_main/Scripts/run_models.py`. This meant keeping the ABT as the full reference file, but applying a narrower set of preparation steps for the training data.

The preparation flow is summarized in Table 16. The biggest change was the restriction to the `open_market` segment. After that, the script removed only rows that could not support the target variable or the spatial-lag feature.

Table 16

Preparation Flow from Frozen ABT to Modeling-Ready Dataset

Preparation Step	Rows Kept	Share of ABT	Rows removed	Re-
Frozen ABT before Chapter 5 filtering	2,047	100.00%	–	
Filter to <code>open_market</code>	1,619	79.09%	428	
Drop three integrity rows	1,616	78.95%	3	
Drop null <code>price_per_sqm</code>	1,509	73.72%	107	
Drop null <code>spatial_lag_price</code>	1,491	72.84%	18	

By the end of this phase, 1,491 rows remained for model fitting. This means 72.84% of the frozen ABT entered the final training workflow. The removed rows were not dropped because they were statistically inconvenient. They were dropped because they could not support the definition of the target variable or the required neighborhood context.

Cleaning and Filtering

The first cleaning step was a scope decision. The ABT kept three market segments for reference: `open_market`, `bank_ropa`, and `floor_price`. However, the deployed task of the thesis was the estimation of open-market residential prices. Because of that, the modeling script filtered the ABT to `market_segment = open_market` before any encoding or model fitting. This removed 428 rows from the broader file and left 1,619 open-market observations.

This step also aligned the training data with the thesis valuation basis. Open-market listings represent the closest available proxy to Market Value in the IVS sense, while bank ROPA and administrative floor-price records reflect different pricing conditions and should not be pooled into the same training target (International Valuation Standards Council, 2025). The ABT still retained those other segments, but they were treated as reference layers rather than training labels.

A second cleaning step handled obvious integrity failures. Three rows with `property_id` values 468, 714, and 769 were removed before model fitting. These rows had implausible unit prices that were not defensible for residential valuation in Metro Cebu. Their removal was a hard data-integrity action, not a general outlier-pruning rule.

The remaining row filters were tied to model requirements. Rows with null `price_per_sqm` were removed because the target variable could not be computed. Rows with null `spatial_lag_price` were also removed because the neighborhood-price feature

was part of the final design. After these two filters, the open-market subset decreased from 1,616 to 1,491 rows.

Property-type cleanup was already applied before this stage. The ABT used a standardized six-class residential taxonomy: Condominium, House and Lot, Townhouse, Single Detached, Vacant Lot, and Apartment. The earlier generic Residential label had already been removed before this stage, so Chapter 5 started from a residential-only file with the cleaned final taxonomy.

Missing Data Treatment

Missing-data treatment followed the practical pattern of the project: keep the file usable, preserve meaning when possible, and avoid inventing information that the sources did not actually provide.

The first structural decision involved size variables. The thesis used `area_sqm` as the main size feature, with `floor_area_sqm` preferred when available and `lot_area_sqm` used as fallback. This was paired with an `is_vacant_lot` flag so that land-only listings stayed visible to the model. This design was important because vacant lots are priced on land area, while most open-market condominium listings do not carry a meaningful lot-area field.

The second issue was structural absence in `lot_area_sqm`. In the open-market segment, `lot_area_sqm` was fully missing because Lamudi listings did not systematically provide lot area. The modeling script treated this as an all-null column, filled it with 0 inside the feature matrix, and then dropped `log_lot_area_sqm` from the OLS matrix because it became a constant field. This kept the matrix complete without pretending that lot area was observed for those listings.

The ABT already carried preparation flags for bedrooms, bathrooms, and floor area. In the open-market segment, the file contained 449 rows with `bedrooms_imputed =`

1, 299 rows with `bathrooms_imputed = 1`, and 107 rows with `floor_area_imputed = 1`. After the final training filters, the remaining 1,491-row modeling subset still contained 397 bedroom-imputed rows and 253 bathroom-imputed rows, while `floor_area_imputed` dropped to 0 because the rows with unresolved floor-area issues were removed together with the null target rows.

The final model-build script still applied one more safeguard step. After filtering and encoding, it checked the remaining numeric columns for nulls and used median imputation where needed. In the current frozen pipeline, this affected bedrooms, bathrooms, `bir_zonal_cr_median`, and `bir_zonal_rc_median`. The all-null `lot_area_sqm` column was handled separately and filled with 0 rather than sent through the median imputer.

Table 17 summarizes the main missing-data actions that carried into the final matrix.

No separate feature scaling or normalization step was applied in this phase. The pipeline did not include a scaler, and the final models did not require one for the implemented workflow.

Market Segment Treatment

The `market_segment` field stayed in the ABT because it was useful for audit and provenance work. It allowed the project to keep `open_market`, `bank_ropa`, and `floor_price` rows in one canonical file. However, it was not used as a model feature in the final training workflow.

Once the dataset was filtered to `open_market` only, `market_segment` became a constant column. A constant string field adds no useful variation, so it was excluded from the feature matrix. This prevented the model from learning a meaningless dummy for a segment that no longer varied in the training data.

Table 17*Main Missing-Data Treatments Used in Chapter 5*

Field or Issue	Treatment Used	Why It Was Needed	Effect on Final Matrix
area_sqm is_vacant_lot	/ Use floor area first, then lot area fallback; keep vacancy flag	Separate space from land-only listings	built land-size signal across property types
lot_area_sqm open_market	in Fill all-null column with 0 in the model script	Lamudi structurally lacked lot-area values	rows Matrix stayed complete; constant log form was dropped from OLS
bedrooms bathrooms	and Keep audit flags in the ABT, then median-impute remaining numeric nulls	Interior details were incomplete in part of the listings	Structural fields stayed usable without dropping many rows
bir_zonal_cr_median and bir_zonal_rc_median	Median-impute remaining numeric nulls after filtering	Some classes were missing by location	zonal Administrative columns stayed usable in the final matrix

This step also kept the model aligned with the actual prediction objective of the Streamlit app. The deployed map estimates open-market residential price levels across Metro Cebu. It does not estimate forced-sale or administrative floor-price values.

Outlier Flagging

The project treated outliers in two different ways. First, the ABT kept a soft `price_outlier_flag` field. This flag marked values outside the 1st to 99th percentile range of `price_php` in the full ABT. In the frozen file, 19 rows had `price_outlier_flag = True`: 14 in `open_market`, four in `bank_ropa`, and one in `floor_price`. These rows were not automatically removed just because they were extreme.

For bank ROPA rows, the same percentile rule was backfilled in the ABT so the flag would be consistent across sources. This preserved the flag as a diagnostic variable without forcing a blanket deletion policy.

Second, the project used a much narrower hard-drop rule for obvious integrity failures. Only the three implausible open-market rows with `property_id` 468, 714, and 769 were removed before model fitting. This distinction mattered. Soft outlier flags were used to warn the analyst that a price was unusual. Hard integrity drops were reserved for rows that were not credible enough to keep in a residential valuation model.

Feature Engineering

Most of the heavy feature engineering had already been completed before this chapter, but Chapter 5 still had to decide which engineered fields would actually enter the model matrices.

The project kept the eight network-distance variables for the tree-based models: Cebu Business Park, Mandaue CBD, Mactan CBD, SRP, Talisay Tabunok, Consolacion,

Naga City, and Airport distances. The `is_mactan_island` flag was also retained in the full matrix because Lapu-Lapu properties sit in a different road-network and bridge-access context.

For accessibility, the final ABT retained the nine category-level Metro Cebu Residential Accessibility Index (MCRAI) fields. The older `amenity_score_*` fields had already been dropped because they duplicated the same concept with a weaker formulation. The ABT also stored `mcrai_composite`, but the model matrix excluded it because it is a weighted summary of the nine component scores and would create redundancy if used together with them.

The `spatial_lag_price` field was treated as a required engineered feature, not an optional add-on. Because of that, rows with null `spatial_lag_price` were removed during the filtering stage instead of being imputed.

The OLS baseline used a narrower version of the feature set than Random Forest and XGBoost. This was done to reduce collinearity and coefficient instability. In the current script, `X_ols` dropped `dist_talisay_tabunok_m`, `is_mactan_island`, `bir_zonal_rr_log`, `is_vacant_lot`, `latitude`, `longitude`, `lot_area_sqm`, `floor_area_sqm`, `area_sqm`, and `log_lot_area_sqm`. It then added `log_floor_area_sqm` as the area term for the log-linear hedonic baseline.

Encoding and Final Modeling-Ready Dataset

The final preparation step converted the cleaned dataset into model matrices. The script one-hot encoded city and property_type using `drop_first = True`. This produced five city dummy variables and five property-type dummy variables in the full feature matrix.

After encoding and numeric-null handling, the full matrix used for Random Forest and XGBoost had shape $1,491 \times 48$. The OLS baseline matrix had shape $1,491 \times 39$ after the extra stability drops described earlier. Table 18 summarizes the result.

Table 18*Final Modeling-Ready Matrices and Split*

Output	Shape or Size	Notes
Full tree-model matrix (X_full)	$1,491 \times 48$	Used for Random Forest and XGBoost
OLS matrix (X_ols)	$1,491 \times 39$	Used for the log-linear baseline after extra stability drops
Training split	1,192 rows	80% of the modeling-ready data
Test split	299 rows	20% of the modeling-ready data
Categorical encoding	5 city dummies, 5 property-type dummies	Created with one-hot encoding and drop_first = True

The final modeling-ready subset remained concentrated in the same main residential markets seen in Chapter 4, especially Lapu-Lapu City and Cebu City. Even after the row drops, the dataset still covered all six LGUs and all six standardized residential property types.

Overall, the data preparation phase was conservative. It did not chase a perfectly tidy dataset. Instead, it applied a small number of defensible filters, preserved the main valuation signals, and made the feature matrix complete enough for model fitting. Chapter 6 discusses how these prepared matrices were used in OLS, Random Forest, and XGBoost.

Modeling

This chapter covers the CRISP-DM modeling phase for the prepared open-market dataset. Chapter 5 described how the training data was assembled. This chapter explains how the three models were specified, fit, and tuned in the current pipeline. Detailed evaluation results are discussed in Chapter 7.

Model Selection Rationale

The study used three models because each one served a different role in the analysis. Ordinary Least Squares (OLS) was used as the interpretable hedonic baseline rooted in implicit-price theory (Rosen, 1974). Random Forest was included because it can capture non-linear relationships and feature interactions without requiring a fixed functional form (Breiman, 2001). XGBoost was included because gradient boosting usually performs well on structured tabular data and was expected to be the strongest predictive model in the final comparison (T. Chen & Guestrin, 2016). Chapter 3 already introduced these models conceptually. The focus here is the actual fitting setup used in the saved implementation.

Train/Test Split

The modeling script took the 1,491-row output from Chapter 5 and split it into training and test subsets using an 80:20 ratio. This produced 1,192 training rows and 299 test rows. The split used `random_state = 42` so that the same rows would be assigned to the same fold every time the script was rerun.

The code also passed `market_segment` to the split function as a stratification variable. In practice, this did not change the final class balance because the modeling dataset had already been filtered to `open_market` only. The stratification argument remained in the script as a carryover from the broader ABT design, but after the open-market restriction it acted only as a consistency check.

All three models were fit on the same target vector, `log_price`. This kept the training target consistent across OLS, Random Forest, and XGBoost. Chapter 7 reports the back-transformed error metrics for comparison.

Table 19

Training Data Used in the Modeling Phase

Item	Value	Notes
Modeling-ready rows	1,491	Output passed forward from Chapter 5
Training rows	1,192	80% of the modeling dataset
Test rows	299	20% of the modeling dataset
Random seed	42	Used for reproducible splitting
Stratification field	<code>market_segment</code>	Constant <code>open_market</code> field after filtering
Common supervised target	<code>log_price</code>	Shared across all three fitted models

OLS Hedonic Baseline

The OLS baseline was fit in *statsmodels* using the `X_ols` matrix and the `log_price` target. This was the most constrained specification among the three models because it had to remain stable enough for coefficient interpretation. The final saved summary in

Models/ols_summary.txt used 1,192 training observations and 39 modeled terms after the constant was added.

The OLS feature set followed three decisions from the modeling log. First, `dist_talisay_tabunok_m` was removed because it was highly collinear with adjacent CBD-distance variables. Second, `is_mactan_island`, `bir_zonal_rr_log`, and `is_vacant_lot` were removed from `X_ols` because they duplicated information already represented by dummy variables or their raw counterparts. Third, the raw area fields were replaced with a log-scale area term so the baseline would follow a log-linear hedonic form rather than a raw linear size effect. In the current script this meant dropping `lot_area_sqm`, `floor_area_sqm`, and `area_sqm` from `X_ols`, then adding `log_floor_area_sqm` back as the retained size term.

These changes were not applied to the tree-based models. Random Forest and XGBoost were allowed to keep the full `X_full` matrix because tree splits are more tolerant of redundant and correlated predictors. The tighter OLS feature set was therefore a stability choice, not a general claim that the dropped columns had no predictive value.

The condition-number check showed why this narrower OLS matrix was necessary. The earlier unstable form had a condition number of about 1.04×10^{16} . The saved OLS summary reported a condition number of 7.98×10^6 . This was still large, but it was much lower than the earlier near-singular specification and was sufficient for the baseline role assigned to OLS in this study.

Random Forest

The production Random Forest model was fit with 300 trees using *scikit-learn*'s *RandomForestRegressor* (Breiman, 2001). The saved object used `n_estimators = 300`, `random_state = 42`, and `n_jobs = -1`. Other controls were left at the library defaults in the current environment. In the saved model object this resolved to settings such as

Table 20*Model Matrices Used in Chapter 6*

Matrix	Shape	Use in the Modeling Pipeline
X_full	$1,491 \times 48$	Full feature matrix used by Random Forest and XGBoost
X_ols	$1,491 \times 39$	Reduced matrix used by the OLS hedonic baseline
OLS target	$1,491 \times 1$	log_price vector fit with <i>statsmodels</i> OLS

max_depth = None, min_samples_split = 2, min_samples_leaf = 1, and max_features = 1.0.

The model was trained on X_full rather than X_ols. This let the ensemble learn from the complete structural, locational, administrative, and accessibility feature set without the stricter collinearity controls imposed on the OLS baseline. No scaling step was applied before fitting.

XGBoost

The baseline XGBoost comparator was fit with the *XGBRegressor* implementation from the *xgboost* package (T. Chen & Guestrin, 2016). The saved baseline object used n_estimators = 300, learning_rate = 0.05, max_depth = 6, random_state = 42, objective = reg:squarederror, and verbosity = 0. Other controls were left at the estimator defaults.

Like the Random Forest model, XGBoost used the full X_full matrix and the same log_price target vector. This kept the comparison fair across the two tree-based models. The baseline configuration was intentionally simple because it already performed well on

the prepared dataset and did not require an extensive parameter stack to produce a competitive fitted model.

Hyperparameter Tuning

Hyperparameter tuning was implemented in a separate script, `thesis_main/Scripts/tune_models.py`. The current confirmation workflow used a 5×3 repeated cross-validation design on the same 1,192-row training subset used by the production pipeline, with RMSE-first refitting so that the saved comparison stayed aligned with the deployment objective in total property price.

For Random Forest, the search was narrowed around the deployed baseline regime through a baseline-centered grid. The best tuned configuration saved by the script used `n_estimators = 400`, `max_depth = None`, `min_samples_split = 2`, `min_samples_leaf = 1`, and `max_features = 0.8`. For XGBoost, the best tuned configuration used `n_estimators = 200`, `learning_rate = 0.07`, `max_depth = 7`, `min_child_weight = 5`, `subsample = 0.9`, and `colsample_bytree = 0.8`.

The tuned models were saved as `rf_tuned.pkl` and `xgb_tuned.pkl`, but they were not retained as production models. When the saved artifacts were evaluated on the held-out 299-row test set, the baseline Random Forest still delivered the strongest overall result with $R^2 = 0.807$. The confirmation-tuned Random Forest reduced MAPE but fell to $R^2 = 0.680$, while baseline and tuned XGBoost reached only $R^2 = 0.492$ and $R^2 = 0.557$, respectively. The baseline Random Forest configuration was therefore retained as the sole production model because none of the tuned or boosted alternatives improved out-of-sample overall performance.

Table 21*Baseline and Tuned Model Configurations Used in Chapter 6*

Model	Key Configuration	Test R^2	Production Status
RF baseline	n_estimators = 300; max_features = 1.0 by current sklearn default	0.807	Retained
RF tuned	n_estimators = 400; max_depth = None; min_samples_split = 2; min_samples_leaf = 1; max_features = 0.8	0.680	Reference only
XGBoost baseline	n_estimators = 300; learning_rate = 0.05; max_depth = 6	0.492	Reference only
XGBoost tuned	n_estimators = 200; learning_rate = 0.07; max_depth = 7; min_child_weight = 5; subsample = 0.9; colsample_bytree = 0.8	0.557	Reference only

MCRAI Stage 2 Weight Derivation

The Stage 2 MCRAI composite was built from the Stage 1 OLS coefficient signs rather than using all nine categories. The final composite did not use all nine MCRAI categories. Instead, it retained only the positive-coefficient OLS categories and renormalized them into a four-part weight vector. The saved weights file in Models/mcrai_stage2_weights.txt reported the final implementation weights as 0.401 for mcrai_education, 0.310 for mcrai_grocery, 0.199 for mcrai_recreation, and 0.102 for mcrai_transport.

Security, tourism, and retail_density were not used in the composite even though

they remained in the model as standalone feature columns. The same was true for finance and health, which stayed in the component set but did not receive positive-composite weights. In other words, the composite index became a narrower summary measure, while the full models still learned from the full set of category-level MCRAI inputs.

An earlier branch of the project had tested SHAP-based weight derivation from the Random Forest model. That approach is recorded in Decisions 14 and 15, but it is now historical only. The controlling specification for the current manuscript is the positive-only OLS composite written to the saved Stage 2 weights file.

Table 22

Final Stage 2 MCRAI Composite Weights

Category	Final Weight	Role in the Current Workflow
mcrai_education	0.401	Retained in composite and as a standalone model feature
mcrai_grocery	0.310	Retained in composite and as a standalone model feature
mcrai_recreation	0.199	Retained in composite and as a standalone model feature
mcrai_transport	0.102	Retained in composite and as a standalone model feature

The current scripts also excluded mcrai_composite from the feature matrices used for model fitting. This avoided rank deficiency because the composite is a weighted summary of the nine component scores that already entered the model separately.

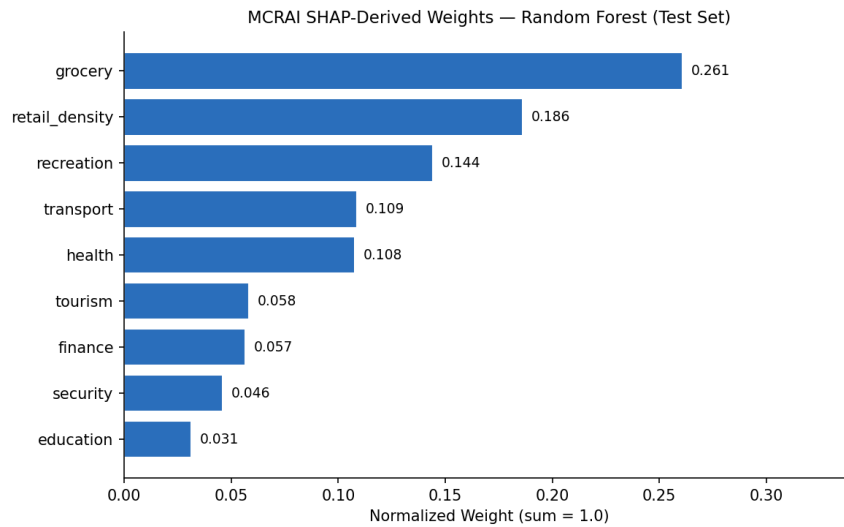


Figure 6

Stage 2 MCRAI Composite Weights Derived from OLS Coefficient Signs. Four positive-coefficient categories were retained: education (0.401), grocery (0.310), recreation (0.199), and transport (0.102). The remaining five categories carried negative or near-zero OLS coefficients and were excluded from the composite.

SHAP Setup

SHAP was configured after model fitting for the two tree-based models using *shap.TreeExplainer* (Lundberg & Lee, 2017). In the current implementation, the explainer was applied to the held-out test set for the saved baseline Random Forest and XGBoost models. The script then computed mean absolute SHAP values by feature and exported beeswarm summary plots to the EDA folder.

This setup was used only for the tree models. The OLS baseline already had an interpretable coefficient structure, so SHAP was not necessary for that part of the pipeline. The SHAP step in Chapter 6 therefore served as the technical bridge between model fitting and the explainability analysis discussed in Chapter 7.

Evaluation

This chapter presents the CRISP-DM evaluation phase for the final set of fitted models. Chapter 6 explained how the OLS baseline, Random Forest, XGBoost, and tuned variants were specified and trained. The focus here is the held-out test-set evidence used to judge model quality, select the production model, and examine the error pattern of the retained model. Broader interpretation of what these results mean for Metro Cebu valuation practice is reserved for Chapter 8.

Evaluation Approach

The study retained four evaluation metrics because each one described model quality from a different angle. All reported values were computed on the 299-row held-out test set after back-transforming predictions from log total property price to total property price in Philippine Peso.

Mean Absolute Percentage Error (MAPE) measured the average percentage gap between the actual and predicted total property price. In plain terms, it showed how far the model missed relative to the size of the observed price. This metric was retained because it is easy to compare across studies and is useful when the thesis needs a single percentage error summary.

Mean Absolute Error (MAE) measured the average absolute peso-denominated error, in this case in total Philippine Peso per property. It showed how large the miss was in the same units used by the back-transformed prediction target. This metric was retained because it is more concrete than MAPE when discussing the size of a typical prediction error.

Root Mean Square Error (RMSE) measured the square root of the average squared error. Because the errors were squared before averaging, RMSE gave more weight to larger misses than MAE. This metric was retained to show whether a model was especially sensitive to a smaller number of very large errors.

The coefficient of determination (R^2) measured how much of the variance in the

observed total property price was explained by the model. It did not describe error size directly, but it remained useful because it showed whether the model captured the broader price pattern in the test set. The study kept R^2 together with MAPE, MAE, and RMSE so that no single metric would dominate the evaluation decision.

Final Model Comparison

Table 23 presents the final comparison across the five evaluated model artifacts on the current post-cleanup sample. The tree-based models still performed much better than the OLS hedonic baseline on the held-out set, but the strongest overall result now came from the baseline Random Forest model, which achieved $R^2 = 0.807$, MAPE = 59.28%, MAE = 4.95 million PHP and RMSE = 27.45 million PHP, all expressed in total property price. The confirmation-tuned Random Forest achieved the lowest MAPE at 53.00%, but that gain came with materially worse R^2 and RMSE. Both XGBoost variants also trailed the baseline Random Forest on the retained overall criteria.

The OLS baseline performed much worse than the tree models on every retained metric. Its held-out R^2 fell to 0.083, while MAPE rose to 201.63%. This gap showed that the interpretable hedonic baseline remained useful as a comparator, but it was not the best predictive option for the current open-market dataset.

The tuned models also failed to displace the retained baseline. Tuning improved XGBoost relative to its baseline and reduced MAPE for Random Forest, but neither tuned variant matched the baseline Random Forest on the held-out RMSE and R^2 balance used for deployment selection.

Best Model Selection

The baseline Random Forest model was selected as the final production model because it had the strongest overall held-out performance under the study's retained

Table 23*Held-Out Test-Set Comparison of Final Model Artifacts*

Model	R^2	MAPE	MAE	RMSE
OLS (log-log baseline)	0.083	201.63%	9.82M	59.82M
Random Forest (baseline)	0.807	59.28%	4.95M	27.45M
Random Forest (tuned)	0.680	53.00%	5.08M	35.34M
XGBoost (baseline)	0.492	60.14%	6.32M	44.54M
XGBoost (tuned)	0.557	58.93%	6.06M	41.58M

multi-metric rule. It achieved the highest R^2 , the lowest MAE, and the lowest RMSE among all evaluated candidates in Table 23. Although the confirmation-tuned Random Forest produced the lowest MAPE, its weaker RMSE and R^2 meant it was not the better deployment choice for total market-value prediction. The Streamlit application is therefore aligned with the saved artifact Models/rf_model.pkl.

The baseline model also performed better than its tuned variants on the retained overall criteria. This suggested that the simpler starting configuration was already close to the useful range for the current training sample of 1,192 rows. The confirmation-tuned Random Forest increased the number of trees and reduced max_features to 0.8, but its held-out performance still remained below the simpler baseline once RMSE and R^2 were considered. The tuned XGBoost also improved over the weaker XGBoost baseline, yet it still generalized materially worse than the retained Random Forest baseline. In this case, tuning did not overturn the incumbent model.

The comparison also showed a clear difference between predictive and explanatory roles. OLS remained the interpretable hedonic benchmark used for coefficient-based reference and for the Stage 2 MCRAI weighting process. The tree models, however, were much stronger at prediction. For that reason, the evaluation phase retained baseline

Random Forest as the production estimator while keeping OLS as the econometric baseline.

This model choice satisfied the predictive part of the thesis objective. The study aimed to build a defensible valuation support model for the open-market residential segment, and the evaluation results showed that the baseline Random Forest artifact was the strongest saved model for that role.

SHAP Feature Attribution

SHAP was applied to the two retained baseline tree models so that their prediction logic could be described in a more transparent way. The ranking used mean absolute SHAP values computed on the held-out test set. Table 24 summarizes the top ten features for each baseline model.

The two baseline models shared a similar top-feature structure. Property-type dummies occupied the top positions in both—property_type_Single Detached and property_type_Condominium ranked first and second in both models, followed by bedrooms and dist_consolacion_m. Size variables (area_sqm and floor_area_sqm), the Mandaue City dummy, the is_vacant_lot flag, and several distance features filled out the remaining top-ten positions, with minor ordering differences between the two models. These common rankings showed that the retained tree models relied on a similar mix of property-type segmentation, structural attributes, and polycentric locational signals.

An important result from this section is what did *not* appear in the top ten. No individual mcrai_* feature entered the top ten list for either baseline tree model. This does not mean that the Metro Cebu Residential Accessibility Index (MCRAI) variables had no role in prediction. It means only that, in the final SHAP ranking for the held-out set, the strongest marginal contributions came first from the distance, coordinate, and property-type variables.

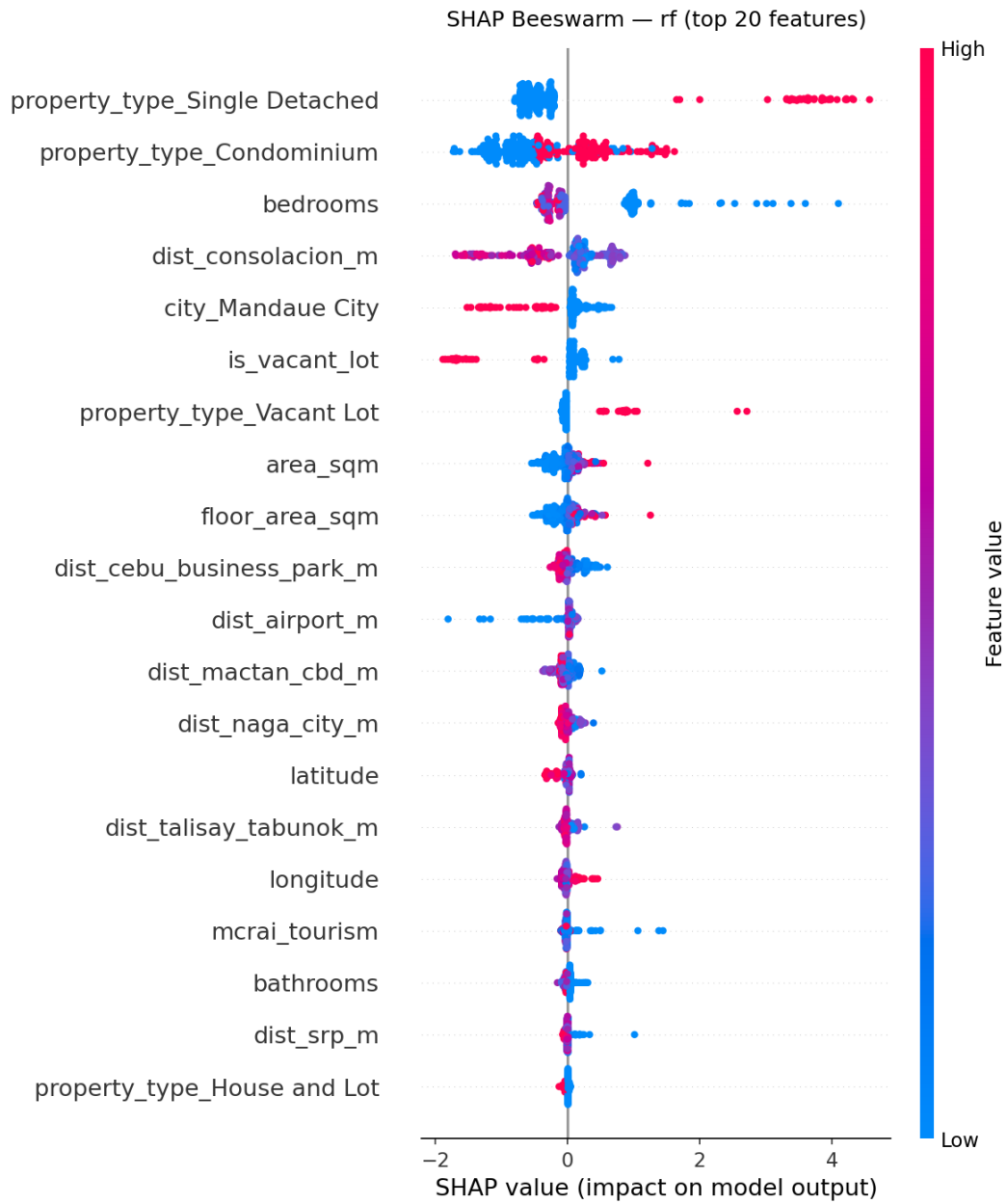
Table 24*Top Ten SHAP Features for the Retained Baseline Tree Models*

Rank	Random Forest	XGBoost
1	property_type_Single Detached	property_type_Single Detached
2	property_type_Condominium	property_type_Condominium
3	bedrooms	bedrooms
4	dist_consolacion_m	dist_consolacion_m
5	city_Mandaue City	floor_area_sqm
6	is_vacant_lot	city_Mandaue City
7	property_type_Vacant Lot	is_vacant_lot
8	area_sqm	longitude
9	floor_area_sqm	dist_naga_city_m
10	dist_cebu_business_park_m	dist_cebu_business_park_m

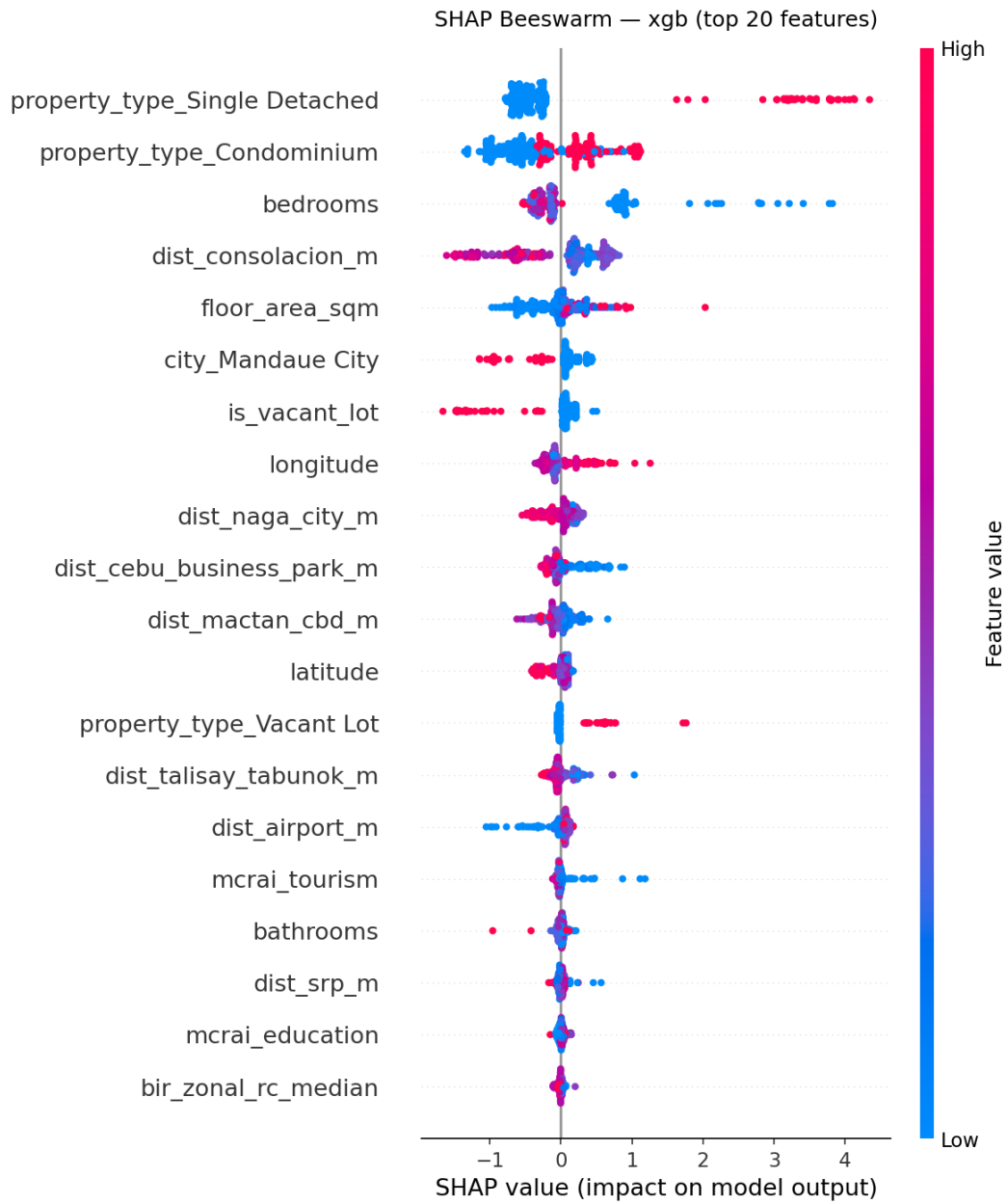
Figure 7 presents the Random Forest SHAP beeswarm plot, while Figure 8 presents the XGBoost SHAP beeswarm plot. These figures were retained as explainability evidence for the final baseline models. The substantive interpretation of why these features dominated in Metro Cebu is deferred to Chapter 8.

Residual Analysis

Residual analysis was centered on the retained production Random Forest model because that was the artifact carried forward to the application layer. The first diagnostic check was the actual-versus-predicted scatter plot shown in Figure 9. The observations followed the expected positive pattern around the 45-degree reference line, but the spread still widened for high-value cases. This was consistent with the RMSE reported in Table

**Figure 7**

SHAP Beeswarm Summary for the Baseline Random Forest Model

**Figure 8**

SHAP Beeswarm Summary for the Baseline XGBoost Model

23. The model captured the overall direction of price variation better than the competing artifacts, but it still produced a number of large misses at the upper end of the market.

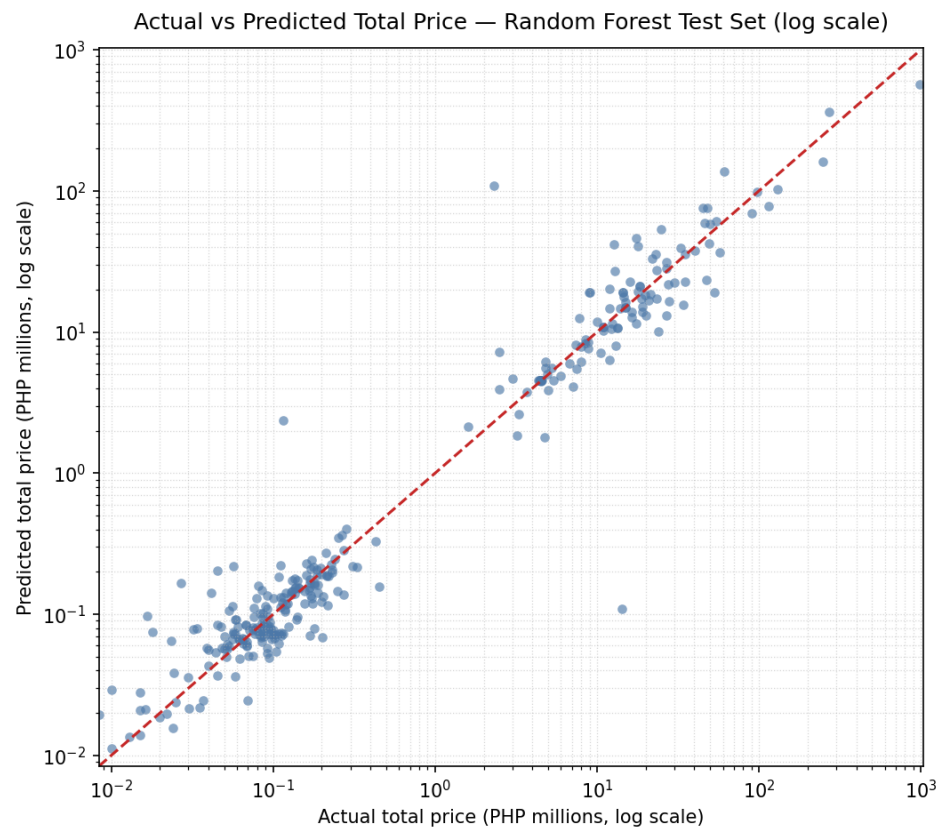


Figure 9

Log-Scale Actual versus Predicted Total Property Price for the Baseline Random Forest Test Set

Figure 10 shows the equivalent scatter for the baseline XGBoost model. The wider spread at the upper end was consistent with XGBoost's weaker RMSE of 44.54 million PHP relative to the retained Random Forest baseline.

No artifact-backed script in the current repository generated reliable residual breakdown tables by LGU or by property type for the held-out test set. For that reason, the residual discussion is limited here to the actual-versus-predicted scatter plot, which is the saved diagnostic figure directly produced in the workflow. That plot is still informative: it shows a positive overall relationship between actual and predicted values, but with visibly

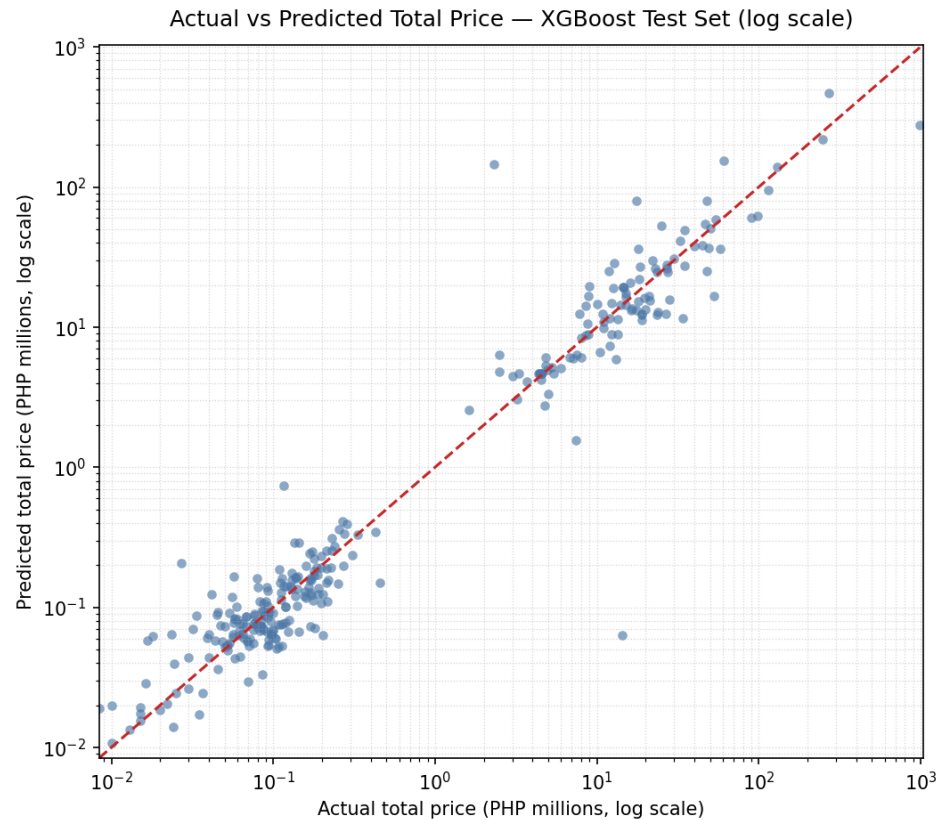


Figure 10

Log-Scale Actual versus Predicted Total Property Price for the Baseline XGBoost Test Set. Compared with the baseline Random Forest, XGBoost showed a wider scatter at the upper end of the price range, consistent with its weaker RMSE of 44.54 million PHP.

wider spread among higher-value cases. In practical terms, the production Random Forest model captured the broad direction of price variation while still producing several large misses at the upper end of the market.

Benchmark Comparison

The production Random Forest result can be placed in context using two studies already discussed in the literature review. Nyanda et al. (2024) reported that boosting achieved a MAPE of 48.0% in the dual-market Dar es Salaam setting, while Random

Forest recorded about 52.7% in the same mixed-market setup (Nyanda et al., 2024). The Metro Cebu deployed baseline Random Forest model posted a higher MAPE at 59.28%, which is worse than both Nyanda et al.'s Random Forest benchmark and their boosting benchmark. The confirmation-tuned Random Forest reduced MAPE to 53.00%, closer to Nyanda et al.'s Random Forest result, but it did so with materially weaker RMSE and R^2 than the deployed baseline. This comparison should be read cautiously because the market structure, price scale, and variable design were different across the studies. Even so, it reinforces that the Metro Cebu listing-based target remains noisy even when the retained model is the strongest overall local artifact.

Sousa et al. (2024) are useful in a different way. Their work did not provide a direct regression benchmark for the present model table, but it supported the use of large online listing datasets to reveal price segmentation that may not be visible in sparse official records (Sousa et al., 2024). That point matters for interpreting the Metro Cebu MAPE. The thesis relied on open-market listing data rather than clean transaction records. As a result, the evaluation used a noisier target that likely carried wider asking-price variation, seller strategy effects, and submarket mixing than a more standardized transaction database.

For that reason, the production model's MAPE should be described honestly. A MAPE of 59.28% was high in absolute terms, especially compared with cleaner Organisation for Economic Co-operation and Development-style automated valuation settings or tightly controlled institutional datasets. At the same time, the result remained defensible for a noisy Philippine listing-based dataset that covered multiple LGUs, multiple residential property types, and a broad open-market price range. The evaluation phase therefore supported a balanced conclusion: the retained Random Forest model was the best available artifact in the current workflow on the retained overall criteria, but it was still subject to substantial prediction error that should be acknowledged clearly in the thesis.

Results and Discussion

Chapter 7 established which model performed best. This chapter shifts to interpretation. The point is not to repeat the evaluation table, but to explain what the retained findings mean for residential valuation in Metro Cebu. Read as a whole, the results point to a market that is polycentric, strongly segmented by property type, and only partly captured by simple counts of nearby amenities.

Overview of Findings

Three points stand out. First, the strongest locational signals were not limited to the old metropolitan core. Property-type dummies occupied the top positions in the tree-based SHAP rankings, with polycentric distance variables following closely. Distance to Consolacion was the strongest locational signal in both models, and distance to Cebu Business Park also appeared in the top ten for both, confirming that the market responds to multiple active nodes rather than one. Second, property type variables occupied a large share of the top explanatory positions, which means that the same location does not generate the same premium across condominiums, detached houses, and vacant lots. Third, the MCRAI results were mixed. Education, grocery, recreation, and transport fed into the final positive composite, while security, tourism, and retail density carried negative OLS signs and were treated as diagnostic rather than additive categories.

Taken together, these findings suggest that Metro Cebu's residential price surface is not organized around one center with a smooth decline outward. It behaves more like a corridor with several active peaks and overlapping catchments. That reading is consistent with the broader polycentric literature and with the local planning view of Metro Cebu as a system of linked urban clusters rather than a single dominant core (Giuliano & Small, 1991; Heikkila et al., 1989; Japan International Cooperation Agency & Metro Cebu Development and Coordination Board, 2015; McMillen, 2003).

Polycentric Structure in the SHAP Top Features

The SHAP rankings are the clearest evidence that the model learned a polycentric rather than monocentric price structure. In both tree-based models, property-type dummies and bedrooms count occupied the top positions, but the locational variables that followed were consistently polycentric. Distance to Consolacion ranked fourth overall in both the Random Forest and XGBoost SHAP tables. Distance to Cebu Business Park appeared in the top ten for both models. Longitude also ranked among the top features in XGBoost, which fits Metro Cebu's north-south coastal form and the east-west separation created by shoreline, bridges, airport land, and mountain terrain.

This does not mean that the traditional core disappeared. In the OLS baseline, distance to Cebu Business Park still carried a significant negative sign. A better reading is that the old core still matters, but it no longer explains the market by itself. The result fits the polycentric logic of Giuliano and Small (1991), the uneven gradient argument of McMillen (2003), and the question raised by Heikkila et al. (1989) about whether subcenters can weaken a single-CBD explanation. In Metro Cebu, the market appears layered rather than centerless.

That interpretation also agrees with the JICA roadmap, which treats Metro Cebu as a linked urban system with specialized clusters and corridor growth nodes such as SRP and Naga (Japan International Cooperation Agency & Metro Cebu Development and Coordination Board, 2015). Distance to Naga City also appeared in the XGBoost top-ten features even though Naga was outside the training LGU set. Its presence as an external anchor in one of the two models suggests that the southern corridor continues to respond to infrastructure and employment signals that extend beyond administrative training boundaries, which is consistent with the roadmap's treatment of Naga as a corridor anchor.

Property Type and the Mactan Island Premium

Property segmentation is part of the main structure of value formation, not a side result. The top SHAP ranks were crowded by property type indicators such as single

detached, condominium, and house and lot. This means the model was learning which residential products absorb location advantages more strongly.

This matters for the Mactan side of the market. The results do not support an exaggerated claim that the island dummy alone drives prices. In the retained SHAP summaries, the island indicator itself did not enter the top 10. At the same time, the broader Mactan signal is clear through the strong role of condominium type, the positive OLS coefficient for Lapu-Lapu City, and the repeated importance of distance to the Mactan CBD. The premium is therefore better read as a submarket effect than as a raw island effect. What the market appears to price is the bundle of airport access, bridge-linked connectivity, tourism-adjacent development, and condominium-oriented demand that characterizes the Mactan corridor.

That reading is consistent with Agosto's finding that mobility and livability matter for Cebu land values (Agosto, 2017). It also fits the JICA roadmap's view of the tri-city core as functionally linked but internally differentiated (Japan International Cooperation Agency & Metro Cebu Development and Coordination Board, 2015). A useful regional comparison is the Bangkok condominium study, which found that transport-related and domestic accessibility factors can explain a large share of condominium value formation in a congested Southeast Asian market (Tochaiwat et al., 2023).

MCRAI Findings and the Spatial Sorting Interpretation

The MCRAI results need a more careful reading than a simple positive-versus-negative split. The final composite retained education, grocery, recreation, and transport because those categories carried positive Stage 1 OLS coefficients in the open-market sample. That result is sensible because these variables describe everyday residential convenience and lower routine household friction.

The negative signs on security, tourism, and retail density should not be read as proof that those categories are inherently harmful. A better interpretation is spatial sorting. In a hedonic framework, households sort across locations according to the

bundles of attributes they prefer and can afford (Bayer & McMillan, 2012; Tiebout, 1956). A variable can therefore carry a negative coefficient not because the amenity itself is undesirable, but because it is concentrated in places with congestion, visitor pressure, commercial intensity, or other traits that push some residential buyers away. The broader capitalization literature supports this conditional reading (Brasington & Parent, 2024; W. Y. Chen & Jim, 2010; Dronyk-Trosper, 2017; Song & Knaap, 2004; Yang et al., 2016).

This is why the Stage 2 MCRAI weight derivation was important. The thesis did not force all categories into one additive score after the Stage 1 signs turned out to be mixed. Instead, it retained the positive household-access categories for the Stage 2 composite and treated the negative categories as diagnostic evidence of sorting and threshold effects. That also lines up with the Metro Manila hedonic report of Tiglao and Rivera and with the Bangkok condominium study, both of which show that accessibility matters, but that the market rewards specific forms of accessibility rather than all forms equally (Tiglao & Rivera, 2006; Tochaiwat et al., 2023).

Comparison to Prior Cebu and Manila Hedonic Literature

The findings are broadly consistent with the earlier Cebu literature, but they refine it. Agosto's conclusion that mobility and livability matter is still visible in the present model (Agosto, 2017). Transport survived into the final MCRAI composite, and the positive categories mostly describe the daily convenience side of livability. What the current model adds is a stronger spatial reading: this capitalization does not operate evenly across the metropolis, but through several active nodes and property-specific submarkets.

The Metro Manila comparison points in the same direction. Tiglao and Rivera treated accessibility to work and urban opportunities as central to both land valuation and location choice (Tiglao & Rivera, 2006). The Metro Cebu results support that idea, but they also suggest that accessibility in a polycentric region is not a single measure. It is a set of overlapping access logics: access to the old core, access to corridor nodes, access to airport-linked condominium markets, and access to daily neighborhood services. In that

sense, Metro Cebu sits between a simple monocentric model and the stronger polycentric claim associated with Heikkila et al. (1989).

A regional comparison also helps explain why the MCRAI categories did not all move in the same direction. The Bangkok condominium article reported that transport accessibility and related factors could account for 51.69% of total model weight in that market (Tochaiwat et al., 2023). The present study found the same broad importance of accessibility, but in a more selective way because the Metro Cebu model covered several residential property types and a wider mix of corridor conditions.

Model Trade-offs

The retained baseline Random Forest model was the best predictive choice under the study's overall evaluation rule, but the findings show why model selection in this thesis could not be treated as a pure one-metric contest. OLS remained useful because it exposed coefficient direction, supported the Stage 1 MCRAI screening, and showed that Cebu Business Park still kept a meaningful role in the price structure. The tree-based models were needed because the market contains non-linear interactions across distance, corridor position, and property type that a single global equation cannot represent well.

The tuned models in Chapter 7 also matter for interpretation. Their weaker out-of-sample performance suggests that the Metro Cebu sample did not reward added complexity by default. The simpler baseline tree models generalized better. That is consistent with recent housing-prediction work that treats tree ensembles as strong tabular learners while still warning that data quality and market heterogeneity remain the real constraints (Nyanda et al., 2024; Sousa et al., 2024).

Limitations

Several limits should stay visible when reading these findings. The model was trained on open-market listings rather than transaction records, so the target variable reflects asking-price behavior as well as market value expectations. The sample was also limited to six LGUs in the training set even though the price surface clearly responds to

nearby external nodes such as Naga City. In addition, the MCRAI categories summarize accessibility through carefully designed but still simplified proxies. They cannot fully represent route quality, travel time variability, perceived safety, or institutional quality.

There is also a methodological limit in the way spatial heterogeneity was handled. The tree-based models captured non-linearity and interaction effects, but the thesis did not estimate spatially varying coefficients directly. That matters in a polycentric region because the value of distance to a node is unlikely to be constant across all locations. The polycentric literature suggests exactly this kind of uneven gradient structure (Heikkila et al., 1989; McMillen, 2003).

Even with these limits, the main interpretation is stable. Metro Cebu's residential valuation pattern is best read as a polycentric market shaped by multiple active centers, property-type segmentation, and selective capitalization of accessibility. Chapter 9 builds on this interpretation by drawing the main conclusions of the thesis, while Chapter 10 turns those conclusions into the study's formal recommendations.

Conclusions

Summary of the Study

This study examined how residential property values in Metro Cebu could be estimated more consistently using a data-driven workflow with geospatial feature engineering. The problem began with the gap between fast-moving market prices and slower administrative or appraisal benchmarks. To address that problem, the study assembled a cleaned analytical base table from open-market listings, bank ROPA inventory, floor-price references, BIR zonal values, and geospatial layers derived from Google Maps and OpenStreetMap. The final deployed prediction scope was limited to open-market residential property in six LGUs: Cebu City, Mandaue City, Lapu-Lapu City, Talisay City, Minglanilla, and Consolacion.

The study followed the CRISP-DM process and compared three modeling approaches: OLS hedonic regression, Random Forest, and XGBoost. The workflow also

constructed a Metro Cebu Residential Accessibility Index (MCRAI) using category-specific radii and gravity-style decay, then paired that index with polycentric distance features, structural property variables, and spatial context variables. The held-out evaluation results showed that the baseline Random Forest model produced the strongest overall performance on the retained deployment criteria, with $R^2 = 0.807$, MAE = 4.95 million PHP, and RMSE = 27.45 million PHP on the test set. The thesis therefore concluded that a machine-learning workflow grounded in local urban structure can produce a usable decision-support tool for Metro Cebu open-market residential valuation, even if prediction error remains substantial in some property types and locations.

Answers to the Research Questions

Research Question 1 asked, “What value drivers significantly influence property prices in Metro Cebu?” The study found that both locational and property-type variables were important, but the strongest recurring drivers were geospatial rather than purely structural. In the retained tree models, property-type indicators such as single detached, condominium, and vacant lot occupied the top positions. Distance to Consolacion was the strongest locational distance variable in both models, ranking consistently in the top four features. Distance to Cebu Business Park also appeared in the top ten for both models, while longitude ranked among the top features in XGBoost. The conclusion was that property prices in Metro Cebu were shaped by a combination of polycentric accessibility and housing submarket segmentation, not by one uniform citywide gradient.

Research Question 2 asked, “Which modeling technique—**Hedonic Regression**, **Random Forest**, or **XGBoost**—produces the most accurate valuation out of sample (lowest MAPE)?” The final comparison made that question more nuanced than the original wording suggested. The confirmation-tuned Random Forest achieved the lowest MAPE at 53.00%, but it did so with materially weaker overall fit than the deployed baseline Random Forest. On the retained multi-metric selection rule, the baseline Random Forest produced the strongest overall held-out result on the 299-row test set, with

$R^2 = 0.807$, MAE = 4.95 million PHP, and RMSE = 27.45 million PHP. OLS performed much worse as a predictive model, while both XGBoost variants remained below the retained baseline on the overall deployment criteria. The conclusion was therefore two-part: the lowest MAPE was delivered by the tuned Random Forest, but the strongest overall out-of-sample valuation model in the final workflow was the baseline Random Forest, which was retained as the production model for the Streamlit application.

Research Question 3 asked, “Do **geospatial features**—proximity to economic nodes, amenity density, and spatial autocorrelation—significantly improve model performance compared to structural-only models?” The study answered this question substantially, though not in the form of a standalone structural-only ablation table. The final evidence showed that geospatial variables were central to the fitted models rather than marginal additions. Polycentric distance features dominated SHAP attribution, longitude remained highly influential, and the MCRAI variables contributed to the econometric screening stage that produced the final positive-category composite. These findings supported the conclusion that geospatial features materially improved the explanatory and practical value of the modeling workflow. At the same time, the final manuscript did not retain a separate structural-only benchmark table, so the exact incremental uplift from geospatial features alone was not isolated as a single chapter-level number.

Research Question 4 asked, “How large is the “Valuation Gap” between the model’s data-driven predictions and traditional BIR Zonal Values?” The study answered this question operationally more than statistically. The workflow defined *valuation gap* as the difference between the model-facing price level and the residential BIR zonal benchmark, and this field was retained as a diagnostic output in the analytical base table and spatial outputs. This made the gap observable at the property and map level. However, the final results chapters did not report one metro-wide summary figure for the valuation gap distribution. The conclusion was therefore narrower: the thesis established a defensible way to measure and visualize the gap, but it did not reduce that gap to a single

final statistic for the whole study area.

Methodological Contribution

The methodological contribution of the thesis rests on the way established valuation ideas were adapted to the specific conditions of Metro Cebu. First, the study scoped the production model to open-market residential property rather than forcing a single predictive target across open-market, bank ROPA, and floor-price observations. That decision kept the deployed output aligned with a market-value use case and avoided mixing fundamentally different pricing regimes into the final prediction surface.

Second, the thesis built a polycentric distance feature set instead of relying on one central-business-district distance measure. This mattered because the final findings showed that Metro Cebu behaved more like a corridor with several active nodes than like a purely monocentric market. Third, the thesis introduced the two-stage MCRAI workflow. Stage 1 used OLS signs and coefficients to evaluate the individual accessibility categories. Stage 2 retained only the positive-coefficient household-access categories in the final composite and treated the negative categories as diagnostic evidence of spatial sorting rather than as simple disamenities. Together, these decisions produced a locally grounded valuation workflow that was more specific to Metro Cebu than a generic hedonic template.

Practical Contribution

The practical contribution of the thesis is the decision-support layer that came out of the modeling work. The main deliverable is a Streamlit price surface for open-market residential property in Metro Cebu. This output does not replace formal appraisal judgment, but it gives brokers, appraisers, lenders, and researchers a reproducible spatial estimate of residential market price under a clearly defined market segment.

The thesis also produced QGIS-ready layers that organized the model outputs spatially. These layers support map-based review of predicted price levels and related diagnostics across the study area. Within that workflow, the *valuation gap* field remains useful because it allows users to compare modeled or observed market-oriented prices

against BIR zonal benchmarks without treating the benchmark itself as the prediction target. In that sense, the thesis contributes not only a model, but also a practical way to inspect where official benchmarks and market-facing estimates may diverge.

Closing Statement

The thesis concluded that data science methods can support residential valuation in Metro Cebu when they are applied with clear market definitions, local geospatial logic, and disciplined treatment of value concepts. The strongest results did not come from model complexity alone. They came from matching the workflow to the structure of the local market: an open-market target, a polycentric urban form, and an accessibility framework designed for household valuation rather than for liquidation or administrative pricing. Under those conditions, the study produced a defensible decision-support tool for Metro Cebu residential valuation.

Recommendations

Recommendations for Valuation Practice

The first recommendation is to use the open-market price surface as a triangulation tool rather than as a replacement for existing valuation references. The final workflow showed that the deployed Random Forest model was the strongest overall predictive option in the study, but Chapter 7 also showed that the held-out error remained substantial in some locations and property types. For that reason, practitioners should read the price surface together with BIR zonal values, bank reserve prices, and property-specific comparable evidence instead of treating the model output as a stand-alone final appraisal.

The second recommendation is to focus comparable-property analysis on the variables that carried the strongest predictive weight in the final models. Chapters 7 and 8 showed that polycentric distance variables and property type mattered more consistently than the accessibility composite alone. In practice, this means valuation review should give more attention to corridor position, proximity to active subcenters, and housing submarket class before giving secondary weight to broad amenity counts.

The third recommendation is to interpret the spatial sorting result from the OLS Stage 2 model carefully. The negative coefficients on security, tourism, and retail density did not mean that these uses are always harmful to residential value. They pointed to context effects such as congestion, commercial intensity, or neighborhood sorting. Practitioners should therefore avoid assuming that physical proximity to police infrastructure, tourism strips, or dense commercial activity automatically adds residential value without checking the surrounding land-use context.

The fourth recommendation is to use the valuation-gap output as a screening device for institutional review. The thesis did not reduce the valuation gap to one metro-wide statistic, but it did preserve the gap as a diagnostic field in the analytical base table and map outputs. This makes it useful for flagging places where market-facing estimates and official BIR benchmarks appear to diverge enough to justify closer appraisal review.

Recommendations for Policy

The first policy recommendation is to treat the thesis findings as support for continued polycentric planning in Metro Cebu. Chapters 7 and 8 showed that the market did not behave as if Cebu Business Park alone structured residential value. Distance signals from Mactan, Consolacion, and Naga also mattered strongly. For regional planners and local governments, this means that investment outside the historical core should continue to recognize the functional role of secondary and corridor-based subcenters rather than concentrating policy attention on the CBP corridor alone.

The second recommendation is to use MCRAI as a reusable planning support framework, but only within the scope for which it was designed. The thesis built MCRAI for residential valuation, not for nationwide equity mapping and not for liquidation pricing. Even so, its category-specific radii, locally defined accessibility categories, and positive-only composite rule can support LGU-level planning or assessment work that needs a parcel-oriented accessibility backbone. The main caution is that the framework

should be adapted carefully to the target use case instead of being copied as a universal score.

The third recommendation is for public institutions that maintain valuation benchmarks. BIR, LGU assessors, and lending institutions could use the price surface and valuation-gap diagnostics as a monitoring layer for places where official benchmarks are drifting away from current market evidence. This does not mean that the thesis model should replace public schedules of values. It means the model can help identify where recalibration work may be most urgent.

Recommendations for Future Research

The first recommendation for future research is to extend the training scope to Naga City. Chapter 8 showed that distance to Naga City was one of the locational signals in the retained models even though Naga was outside the training LGU set. A future version of the study should collect around 150 to 200 verified open-market Naga listings, pass them through the same geocode and enrichment pipeline, and test Naga as a seventh training LGU.

The second recommendation is to estimate spatial heterogeneity more directly. Chapter 8 already noted that the present models captured non-linearity but did not estimate spatially varying coefficients. Geographically Weighted Regression or Mixed Geographically Weighted Regression is the next logical extension. A follow-up study should test whether the capitalization of Cebu Business Park, Mactan, Consolacion, SRP, and Naga changes materially across submarkets instead of assuming one average distance effect for the whole region.

The third recommendation is to calibrate the accessibility framework more empirically. The decay parameter β was fixed at 2.0 as a defensible baseline, but category-specific or empirically estimated decay parameters would be better than one fixed value across all amenity types. A follow-up study should estimate the decay rate by category and should also continue refining the transport layer so that the network cutoff

and road-accessibility interpretation remain consistent with real travel behavior.

The fourth recommendation is to add nonlinear distance-band features for known externalities that cannot be represented well by standard positive-only accessibility scoring. Gasoline stations were excluded from the accessibility index because the literature points to a negative effect at very close range that a Hansen-style score would misread. The same logic can be tested for very-close tourism corridors, heavy roadside activity, or other locally noisy uses. These variables should be modeled as banded or threshold effects rather than folded into one generic proximity score.

The fifth recommendation is to strengthen the time dimension of the analytical base table. The current ABT remained cross-sectional, which is why BSP RPPI could not be used as a per-row time-trend control. A future version should preserve collection dates at the property level so that macro price drift and listing timing can be modeled more directly.

The sixth recommendation is to verify the evidentiary basis for the tourism externality interpretation before extending it. The negative OLS coefficient on tourism density in the Stage 2 model was interpreted cautiously in Chapter 8, but the international literature on tourism-corridor externalities for residential property is thinner than the literature on other commercial uses. Future researchers who want to extend the tourism-density framing should confirm that supporting studies are retrievable through Scholar or Scopus before building on them.

Recommendations for Deployment

The first deployment recommendation is to keep the Streamlit app tied to the project virtual environment so that all dependencies, including SHAP, are available at runtime. The app should be launched through the project environment rather than the system Python. A simple launcher script or startup note that activates the correct environment before running the app is sufficient for most handoff scenarios.

The second deployment recommendation is to ensure the map layer uses a

tokenless base tile provider. CartoDB Positron or a comparable public tile source avoids any dependency on external API keys and keeps the price surface page functional across different host environments without additional configuration.

The third recommendation is to make the model scope explicit everywhere a user sees the app. The price surface predicts open-market residential levels only. It does not predict bank ROPA liquidation values, government floor prices, or every possible property segment in the ABT. That scope should appear clearly in the interface, documentation, and demo script so users do not read the surface as a universal market-value map.

The fourth recommendation is to formalize a refresh cycle for data and model maintenance. The POI inventory, Lamudi listing coverage, and official benchmark schedules will drift over time. For that reason, the deployment workflow should include a regular update cadence for POI refresh, ABT rebuilding, prediction export, and model retraining. A six-month cycle is a reasonable starting point for keeping the app useful without treating it as a live real-time system.

The fifth recommendation is to keep the Streamlit app and QGIS outputs consistent with each other. Both deliverables draw from the same saved model artifact and the same prediction export. Whenever the model is retrained or the ABT is updated, the prediction layer should be regenerated and verified against both the app and the QGIS map before either is shown externally.

Closing Note

This thesis should be read as a demonstrated workflow more than as a finished valuation product. Its main contribution is that the process is reproducible: a custom MCRAI design, a polycentric distance feature set, and a Random Forest plus SHAP pipeline can be rebuilt, checked, and adapted for other Philippine secondary cities with similar data conditions. What this study finished was a defensible template for local valuation analytics, not the last version that Metro Cebu will ever need.

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Appendix A

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Appendix A: Data Documentation

Table A1

Sample Structure of Cleaned BDO Foreclosure Records

Raw Column (BDO Excel)	Processed Feature	Description
REGION	Region	Filtered to Region VII
CITY_PROVINCE	City	Filtered to Metro Cebu LGUs
PROPERTY_ADDRESS	Address	Geocoded string
LOT_AREA	Lot Area	Numeric (sqm)
FLOOR_AREA	Floor Area	Numeric (sqm)
MINIMUM_BID_PRICE	Actual Price	Numeric target variable (PHP)
PROPERTY_DESCRIPTION	Bedrooms, Bathrooms	Parsed via regex

Table A2*Sample Structure of Cleaned Lamudi Listing Records*

Raw Scrape)	Field (Web	Processed Feature	Description
Location		Barangay, City	Parsed location string
Price		Actual Price	Target variable (PHP), cleaned of currency symbols
Bedrooms		Bedrooms	Numeric extract
Bathrooms		Bathrooms	Numeric extract
Floor area		Floor Area	Numeric extract (sqm)
Land Size		Lot Area	Numeric extract (sqm)
Latitude / Longitude		Latitude, Longitude	Direct coordinate mapping

Table A3*Final Feature Matrix Schema Used for the Modeling Workflow*

Feature Group	Variables	Type
Identifiers	Property ID, source, market segment (stored in	Metadata
Metadata	ABT; excluded from final fitted feature matrix)	
Structural	Lot Area, Floor Area, Bedrooms, Bathrooms, Property Type, vacancy / imputation flags	Numeric / Categorical
Locational	City, Barangay, Latitude, Longitude, <i>is_mactan_island</i>	Numeric / Categorical
Geospatial	Network distance to Cebu Business Park, Mandaue CBD, Mactan CBD, SRP, Talisay Tabunok, Consolacion, Naga City, and Airport	Numeric (meters)
MCRAI	<i>mcrai_education</i> through <i>mcrai_retail_density</i> , plus <i>mcrai_composite</i>	Numeric (index scores)
Administrative / Neighborhood	BIR zonal-value features and <i>spatial_lag_price</i>	Numeric (PHP)
Target	<i>price_per_sqm</i> , <i>price_php</i> , <i>log_price</i>	Numeric

Appendix B

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Appendix B: Supplementary Tables

Table B1*Geographic Distribution of the Final ABT by LGU and Market Segment*

City or Municipal- ity	Rows	Share	open_market	bank_ropa	floor_price
Cebu City	723	35.32%	488	211	24
Lapu-Lapu City	643	31.41%	558	44	41
Mandaue City	351	17.15%	307	11	33
Talisay City	122	5.96%	84	37	1
Consolacion	119	5.81%	111	4	4
Minglanilla	89	4.35%	71	13	5
Total	2,047	100.00%	1,619	320	108